UNIVERSITY of HOUSTON | ECE

ECE 3337: Signals & Systems Analysis

Lecture 3: Signals and Systems

Reference: Chapter 1
(Sections 1.4.4 – 1.5)

Please turn off your gadgets now

Recap: Basic Waveforms

$$x(t) = A\sin\left(\frac{2\pi t}{T_0} + \theta\right)$$

The Sinusoid

$$e^{j\omega_0 t} = \cos(\omega_0 t) + j\sin(\omega_0 t)$$

Exponential representation of sinusoid

$$x(t) = x(t + nT_0)$$

General Periodic Signal

$$x(t) = Ae^{j\omega_0 t} = \left(|A|e^{j\theta}\right)e^{j\omega_0 t} = |A|e^{j(\omega_0 t+\theta)}$$

Complex Exponential

$$u(t) = \begin{cases} 0 \ \text{for } t < 0 \\ 1 \ \text{for } t > 0 \end{cases}$$

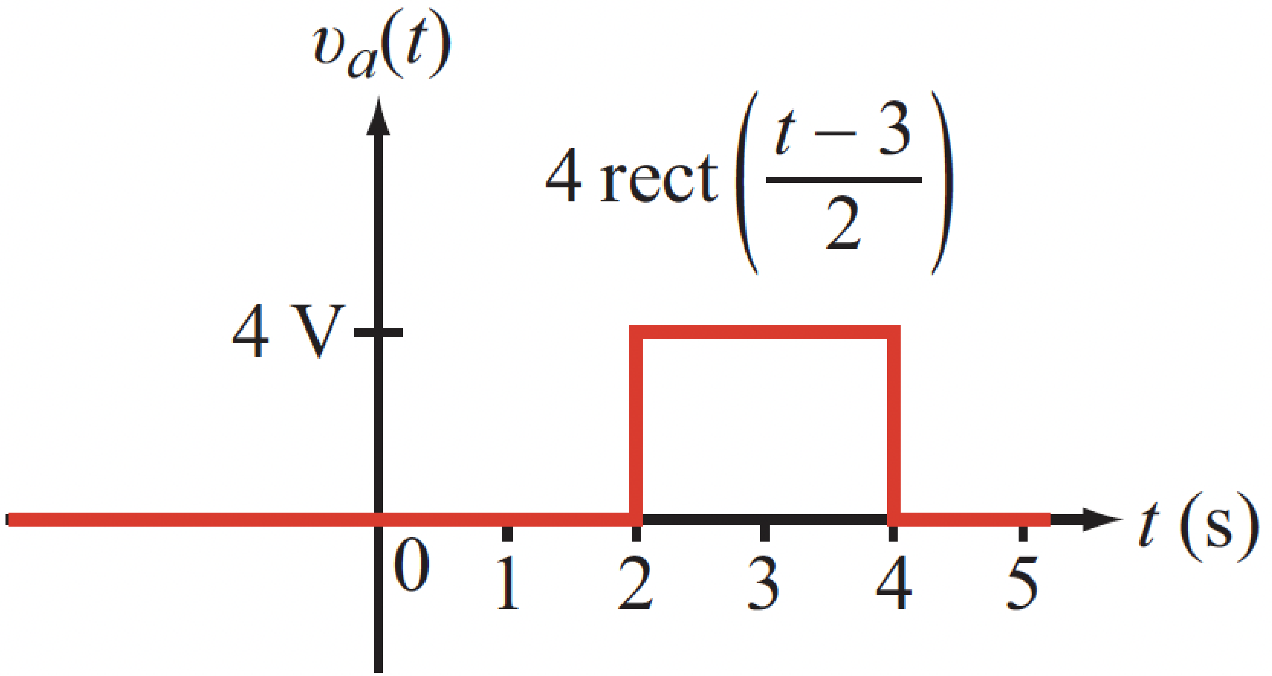
The Unit Step

$$p(t) = u(t) - u(t-1)$$

The Pulse

Unit Step & Pulse: Example

Write a formula for the following waveform

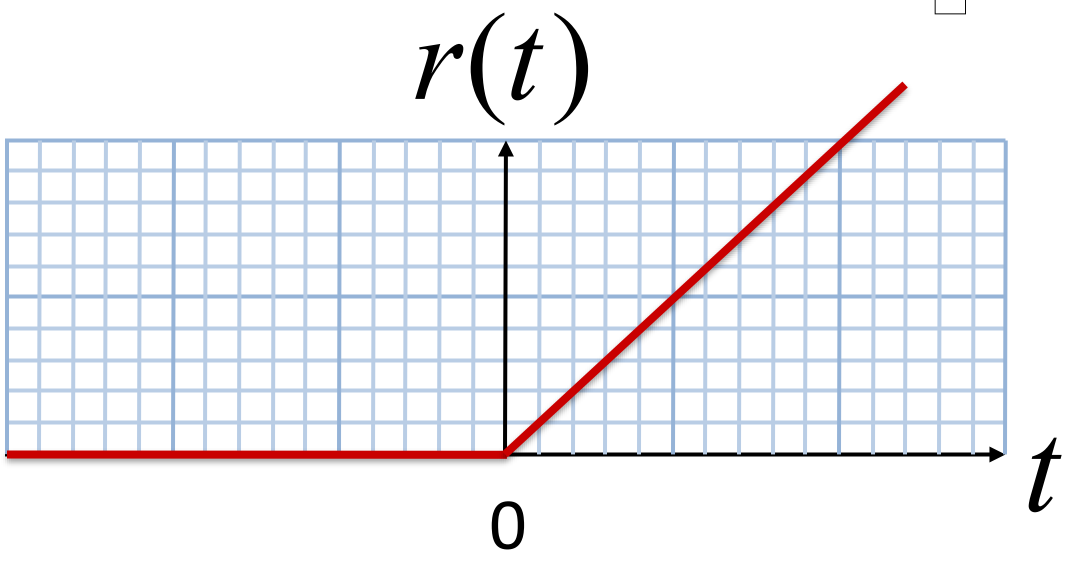


Correct Answer:

$$v_a(t) = 4u(t-2) - 4u(t-4)$$

Integrating the Unit Step: The Ramp

$$\int_{-\infty}^{t} u(\tau)\,d\tau = ?$$



$$r(t) = \begin{cases} t & \text{for } t > 0 \\ 0 & \text{for } t < 0 \end{cases}$$

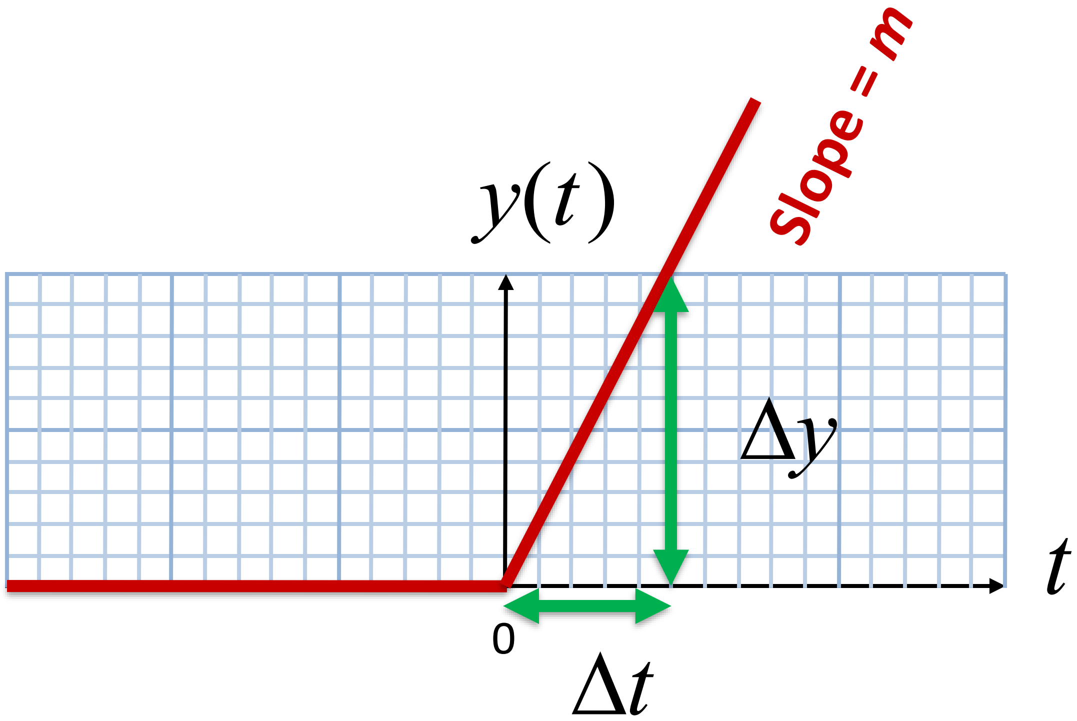
The Ramp Function

$$r(t) = t\,u(t)$$

$$\frac{dr(t)}{dt} = \begin{cases} 0, & t < 0 \\ 1, & t > 0 \end{cases}$$

Slope = 1

The Scaled Ramp



$$m = \frac{\Delta y}{\Delta t}$$ **Slope**

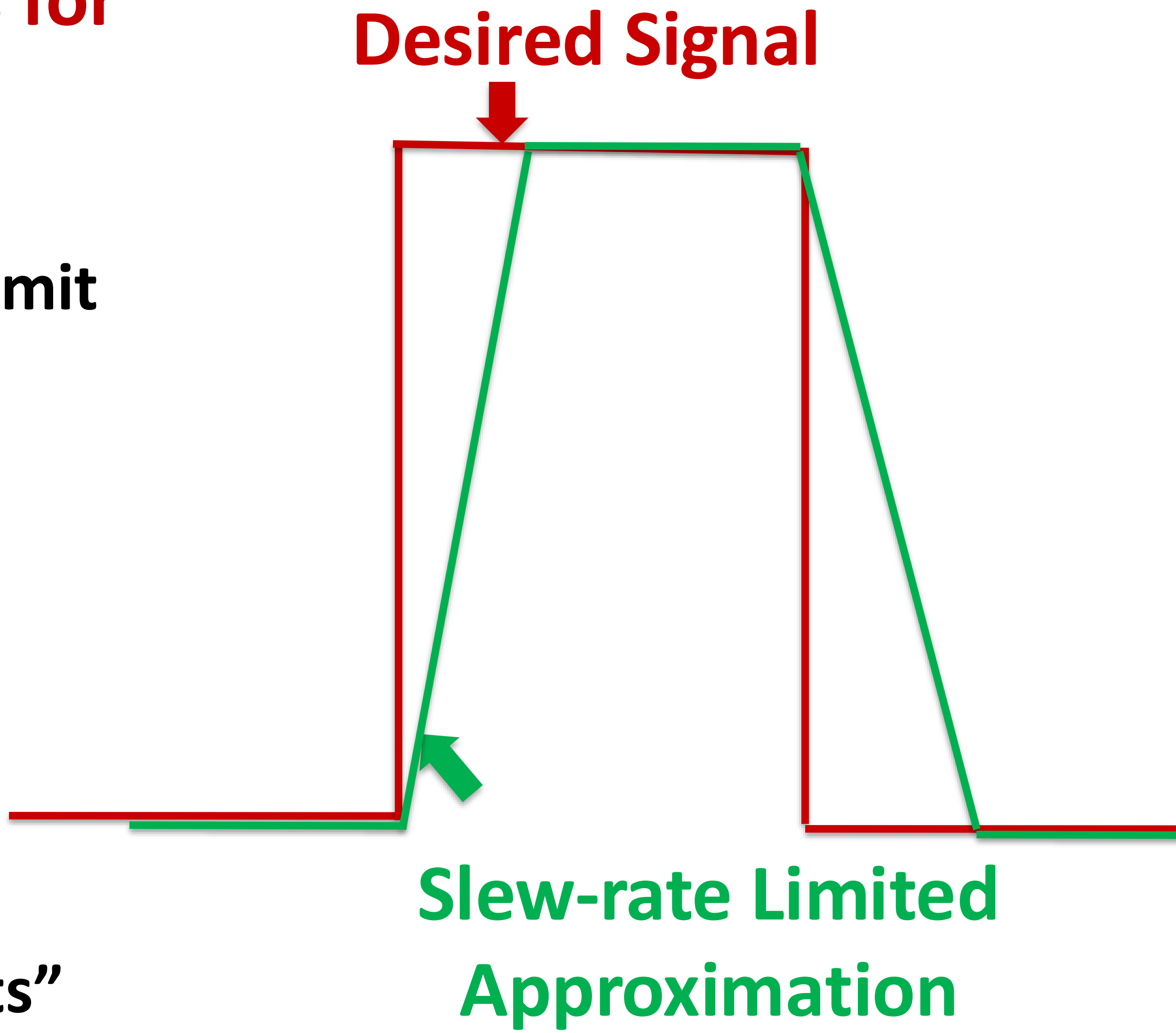
$$y(t) = m\,r(t) = m\,t\,u(t) = \int_{-\infty}^{t} m\,u(\tau)\,d\tau$$

Why Care about Ramps?

Ramps are versatile building blocks for representing real signals

– Real circuits cannot vary voltages/ currents faster than a "slew rate" limit

$$\text{Slew Rate} = \max\left|\frac{d\vartheta(t)}{dt}\right|$$

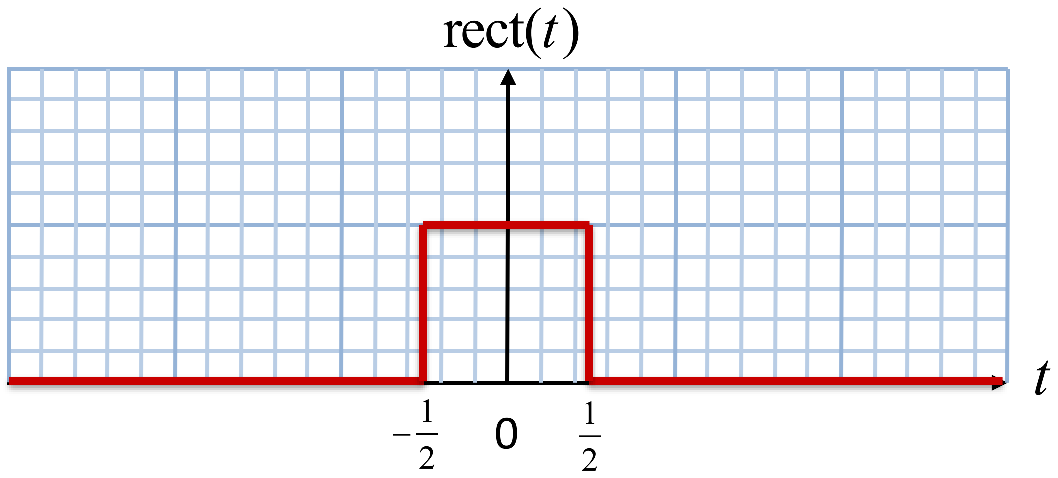


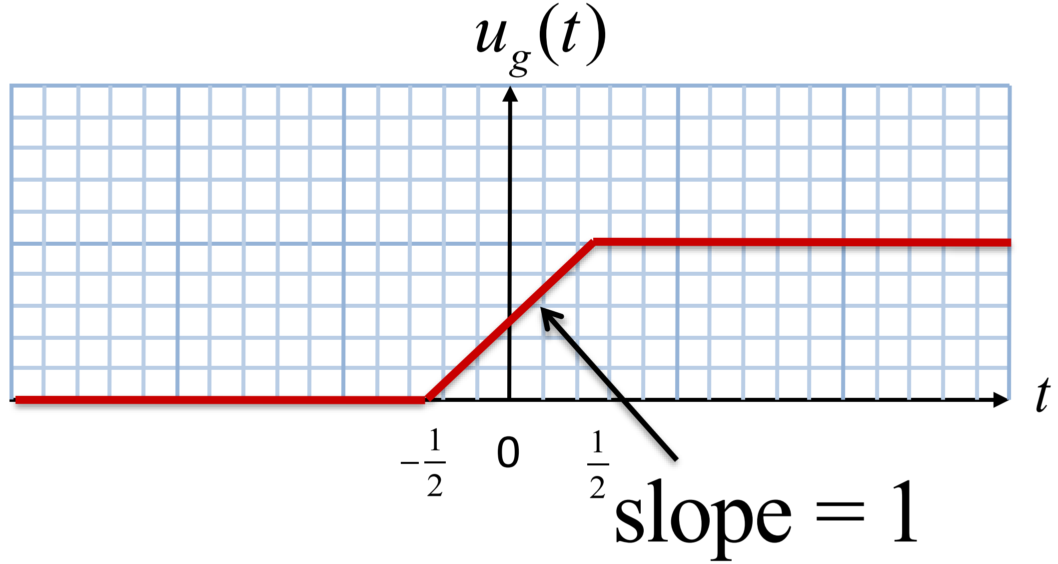
Ramps are also useful for handling many non-signals

– When we get to Chapter 6, we will need ramps for creating "Bode Plots"

The Gradual Step Function

Let’s integrate the rectangle function:



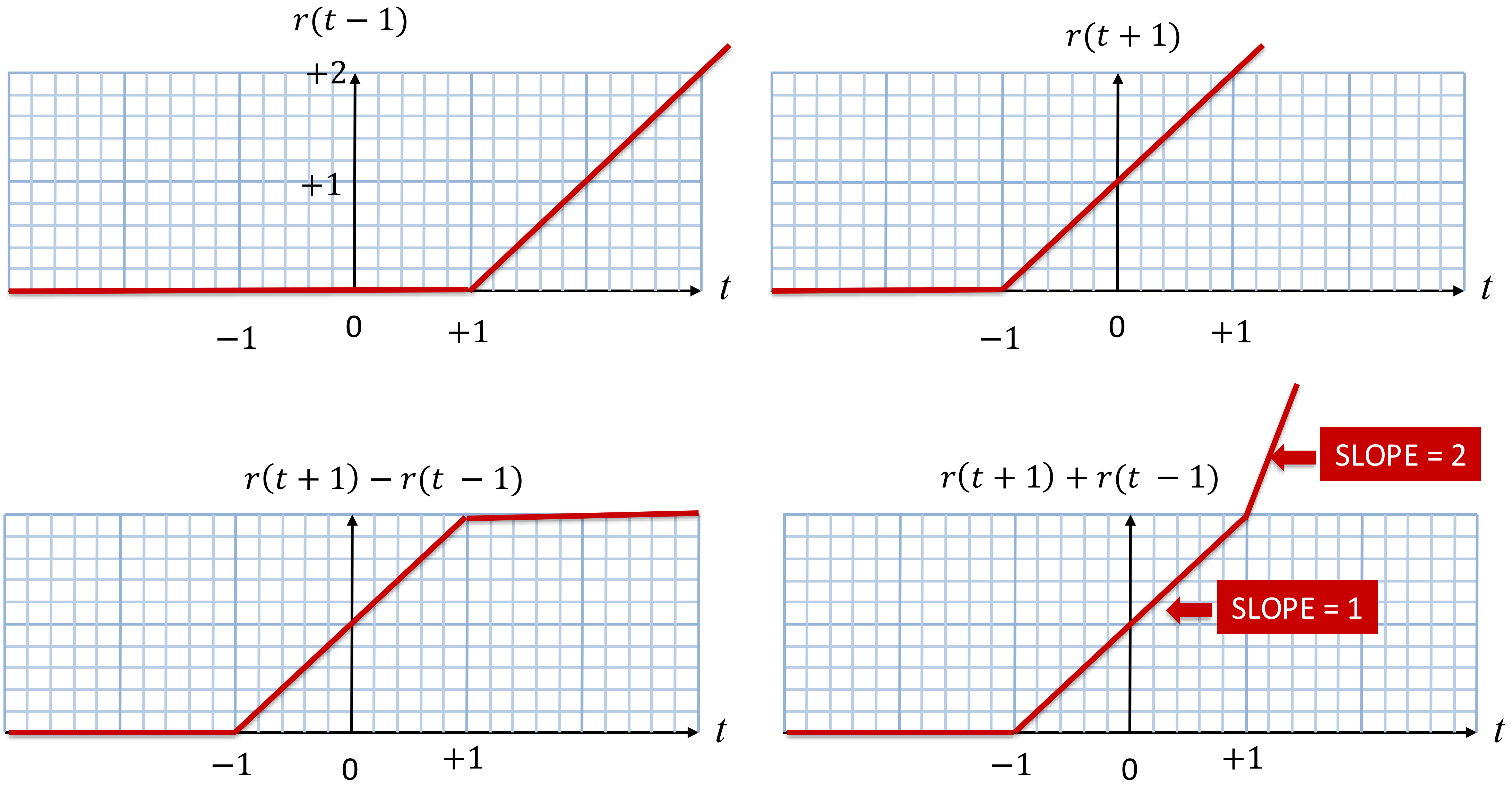


$$u_g(t) = \int_{-\infty}^{t} \mathrm{rect}(\tau)\,d\tau$$

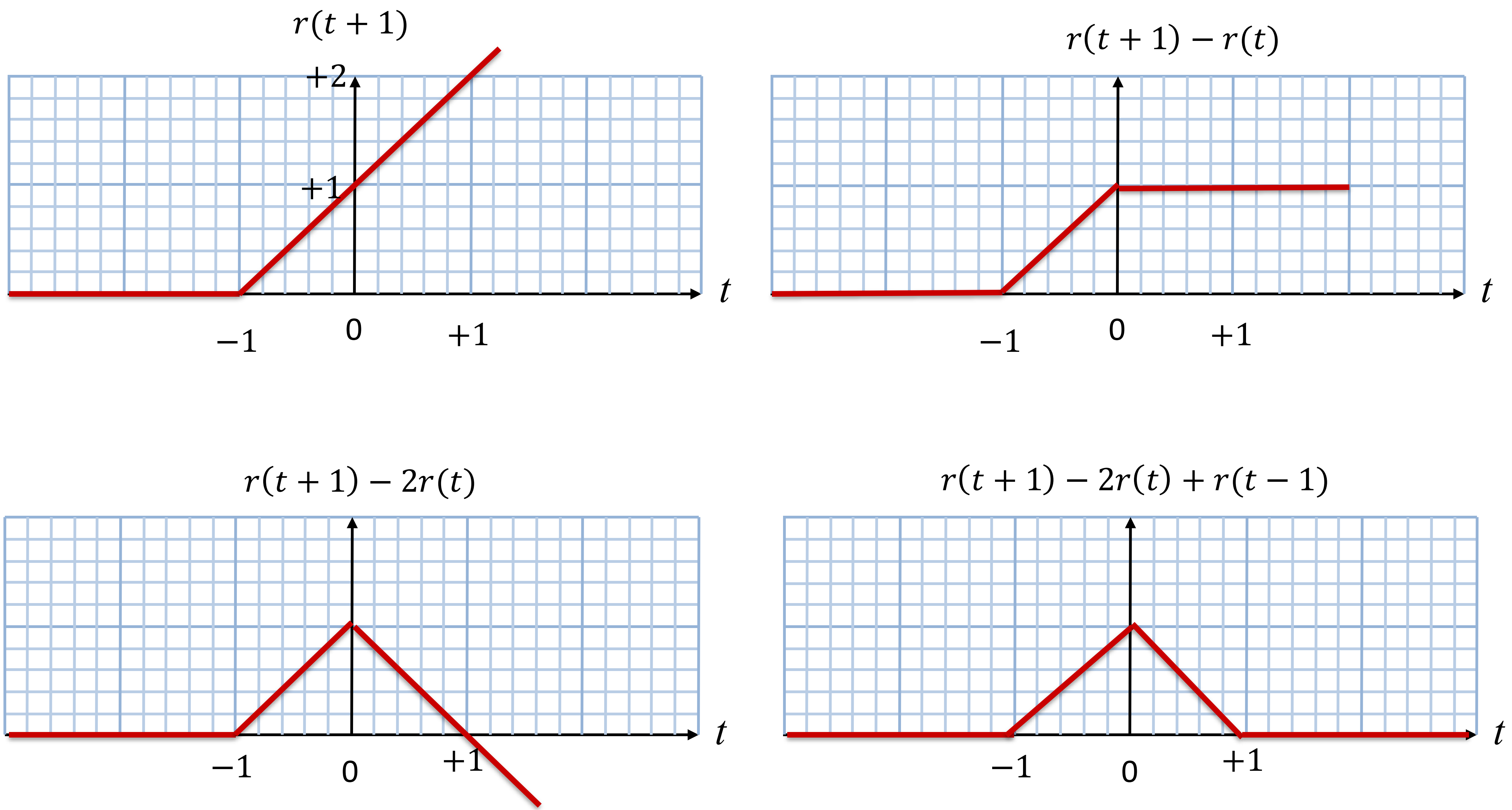
The Gradual Step Function

$$u_g(t) = r\left(t + \frac{1}{2}\right) - r\left(t - \frac{1}{2}\right)$$

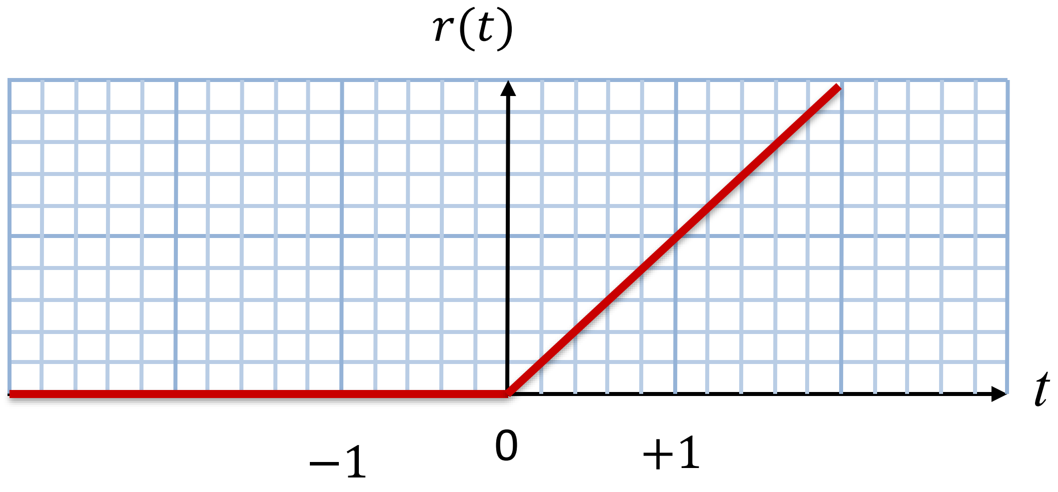
Ramp Examples

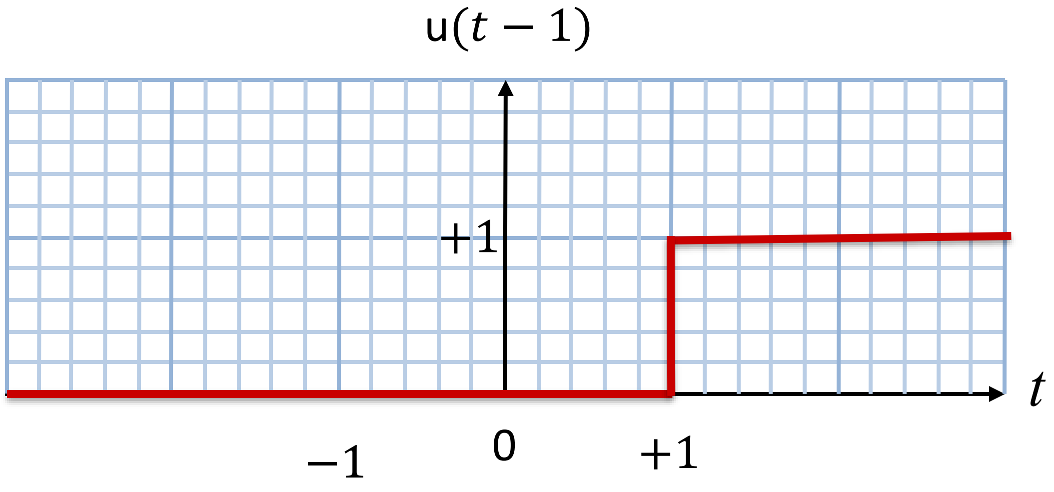


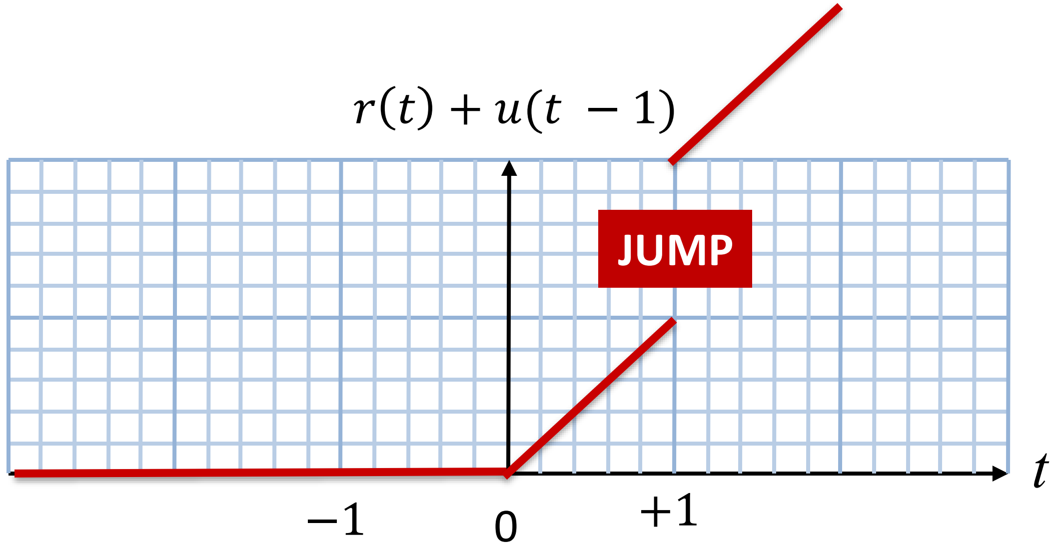
More Ramp Examples

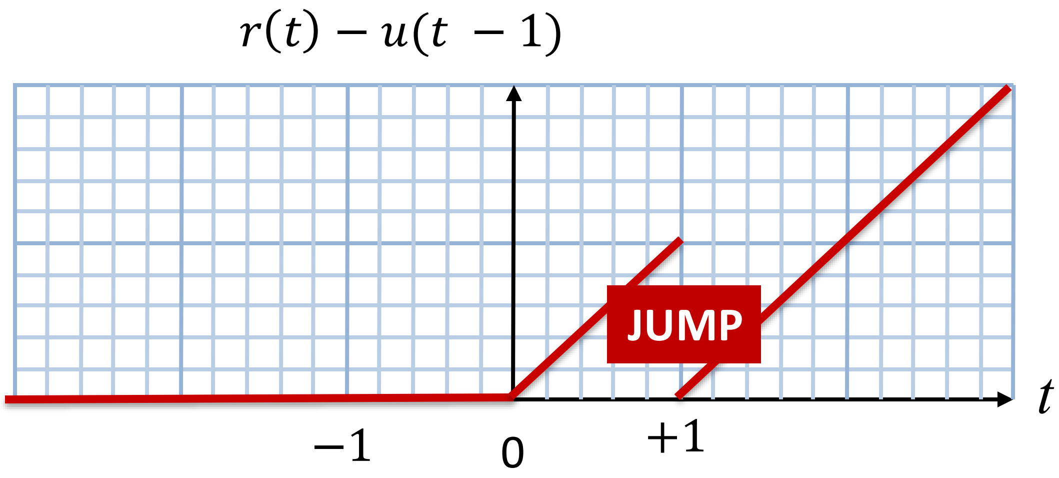


Ramps and Steps



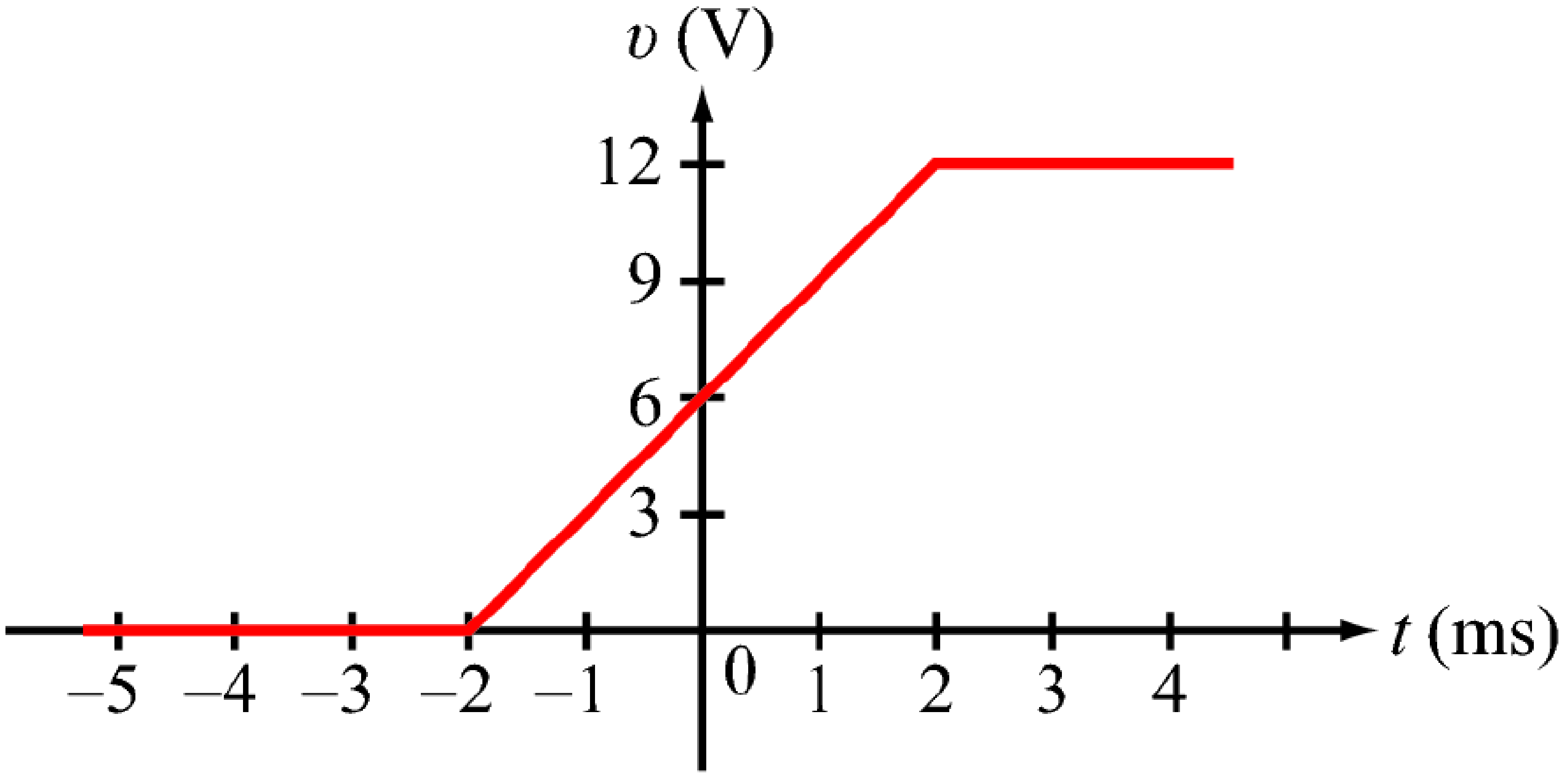






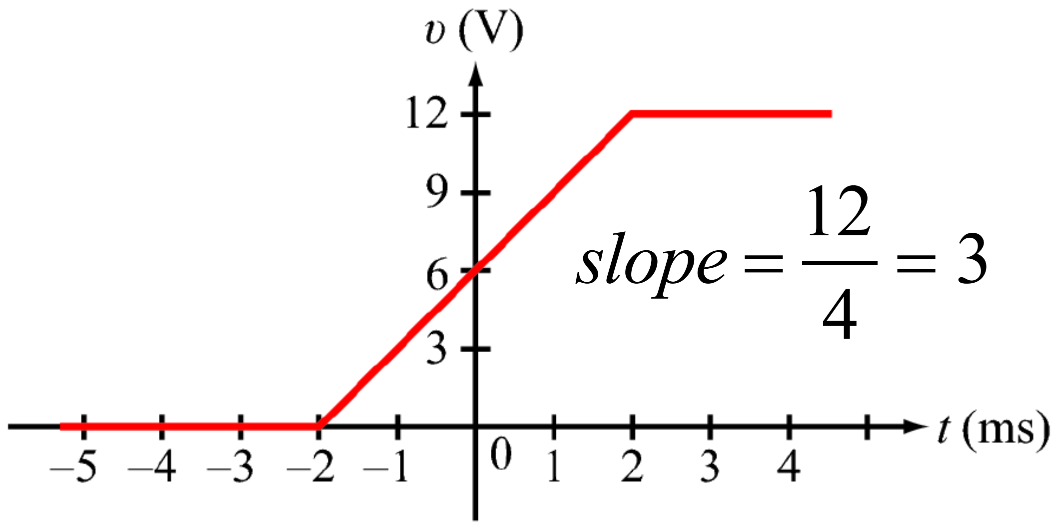
Signal Synthesis with Ramps

Represent the following gradual step function in terms of ramp functions



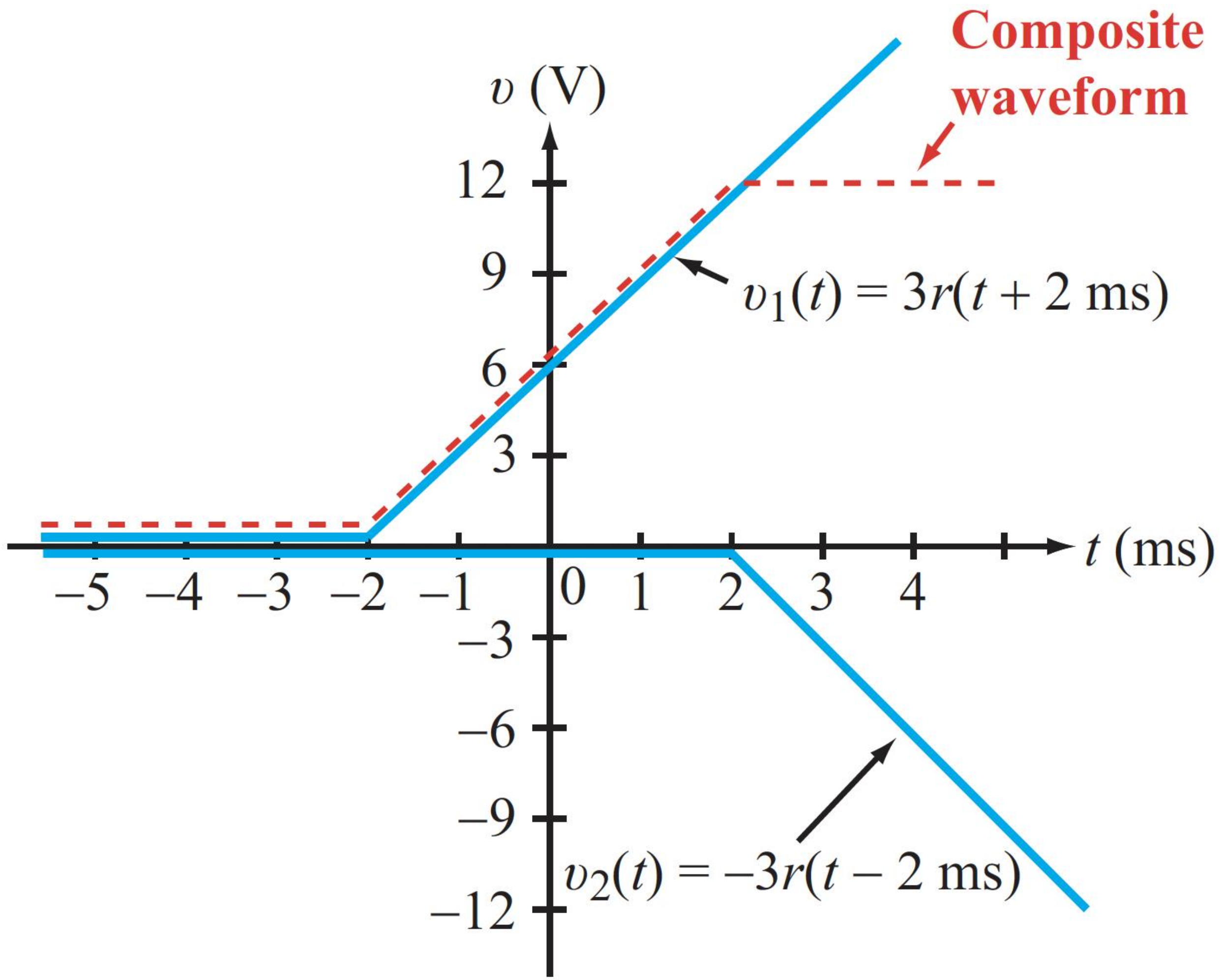
Signal Synthesis with Ramps

Represent the following gradual step function in terms of ramp functions



$$slope = \frac{12}{4} = 3$$

$$\upsilon(t) = 3r(t+2) - 3r(t-2)$$



Examples

$$y_1(t) = 2r(t+1)$$

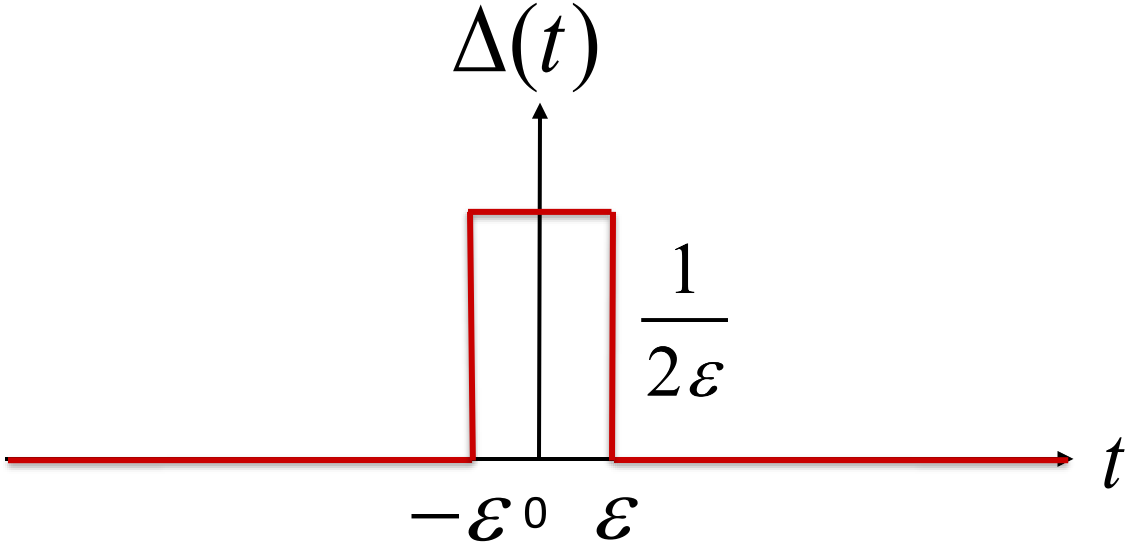
$$y_2(t) = 2r(t-1)$$

$$y_3(t) = 2r(t) - r(t-1)$$

$$y_4(t) = r(t) + 2u(t-1)$$

The Rectangle with Fixed Area

Set the height (amplitude) equal to the width, so the area is always 1



$$\text{width} = 2\varepsilon$$

$$\text{height} = \frac{1}{2\varepsilon}$$

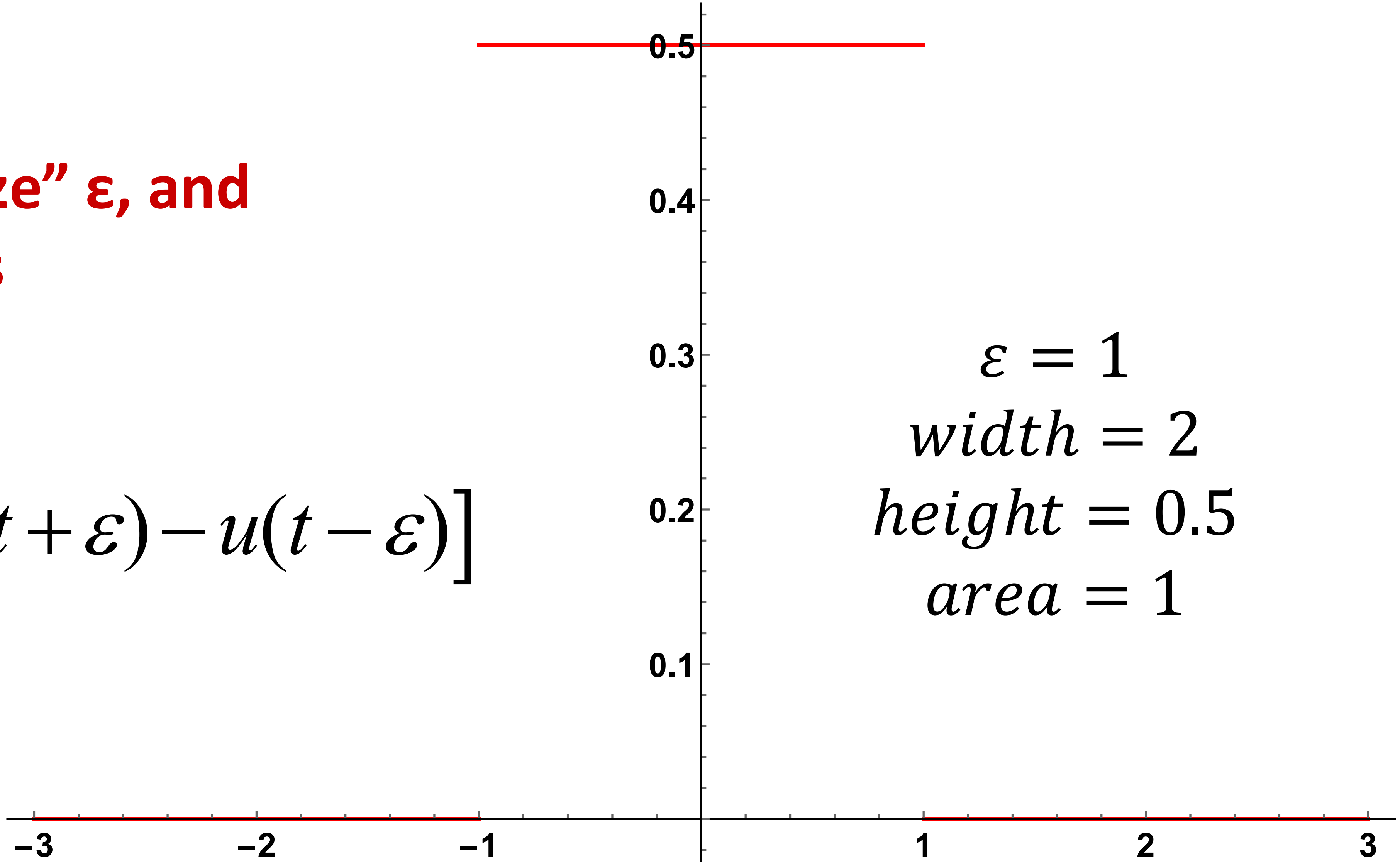
$$\text{area} = 2\varepsilon \cdot \frac{1}{2\varepsilon} = 1$$

$$\Delta(t) = \frac{1}{2\varepsilon} \cdot \left[u(t+\varepsilon) - u(t-\varepsilon)\right] = \frac{1}{2\varepsilon} \cdot \text{rect}\left(\frac{t}{2\varepsilon}\right)$$

The Rectangle with Fixed Area

Now, let's "squeeze" ε, and see what happens

$$\Delta(t) = \frac{1}{2\varepsilon} \cdot \left[u(t+\varepsilon) - u(t-\varepsilon) \right]$$

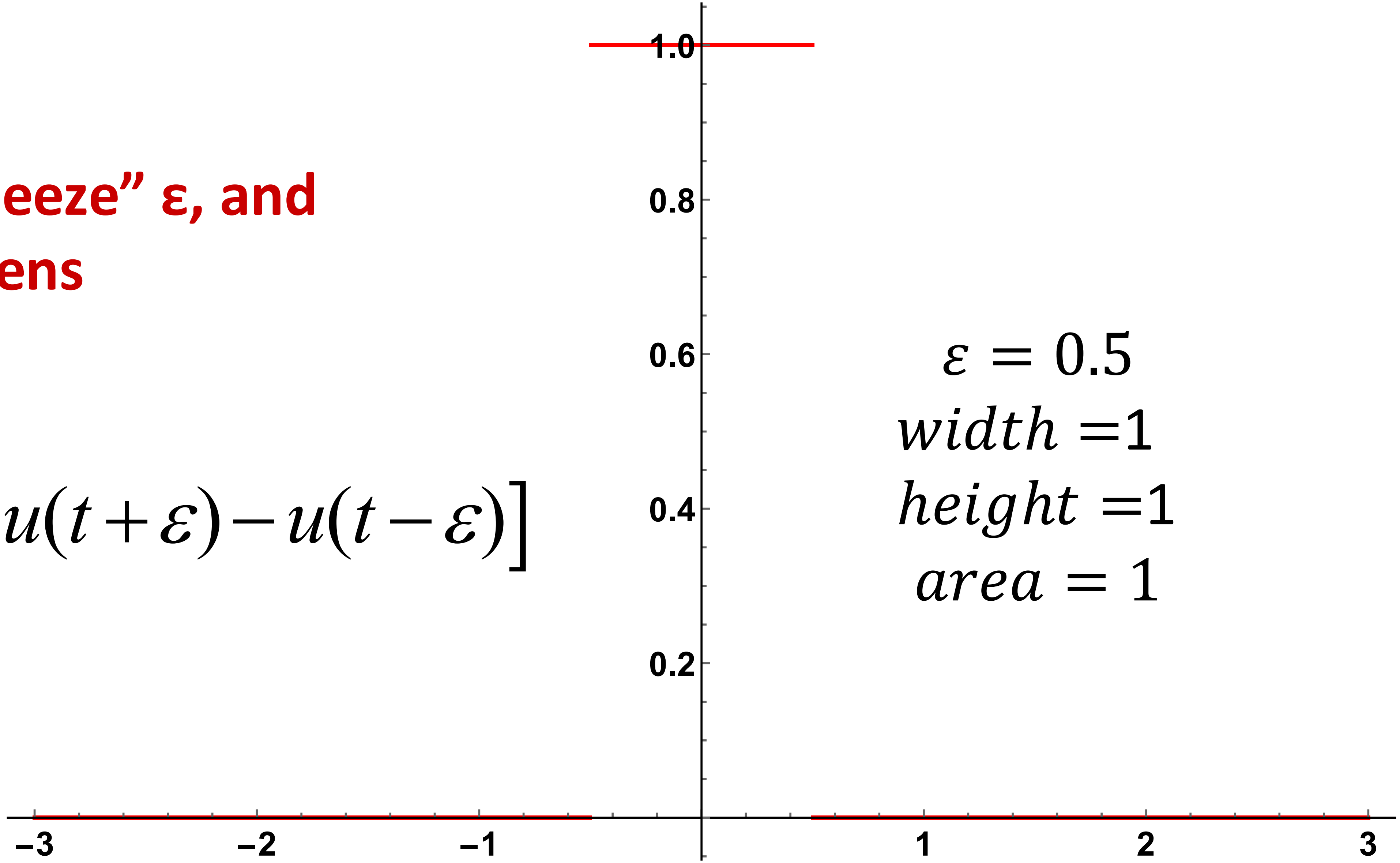


$$\varepsilon = 1$$
$$width = 2$$
$$height = 0.5$$
$$area = 1$$

The Rectangle with Fixed Area

Now, let's "squeeze" ε, and see what happens

$$\Delta(t)=\frac{1}{2\varepsilon}\cdot\left[u(t+\varepsilon)-u(t-\varepsilon)\right]$$



$\varepsilon = 0.5$

$width = 1$

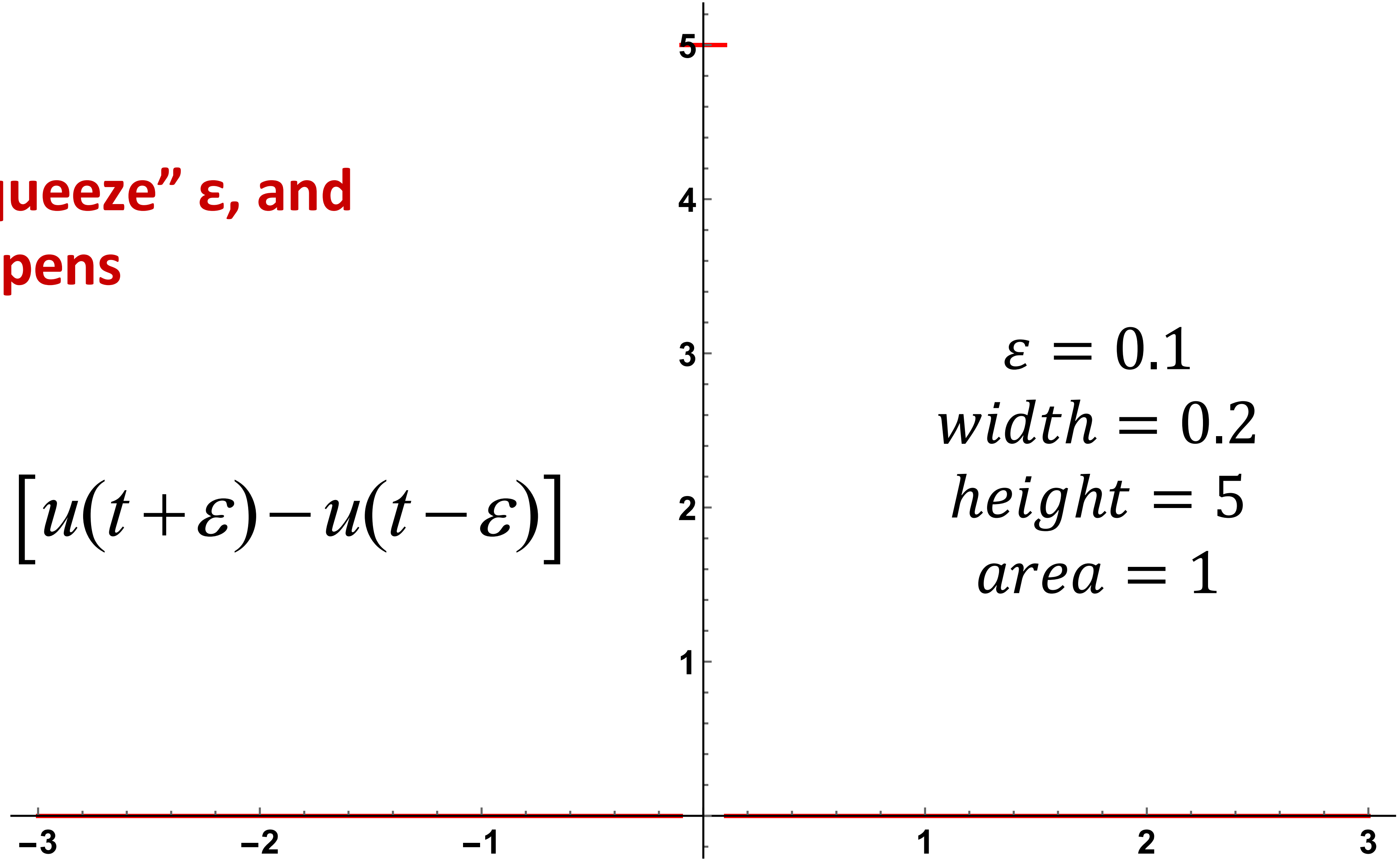
$height = 1$

$area = 1$

The Rectangle with Fixed Area

Now, let's "squeeze" ε, and see what happens

$$\Delta(t) = \frac{1}{2\varepsilon} \cdot \left[u(t+\varepsilon) - u(t-\varepsilon) \right]$$

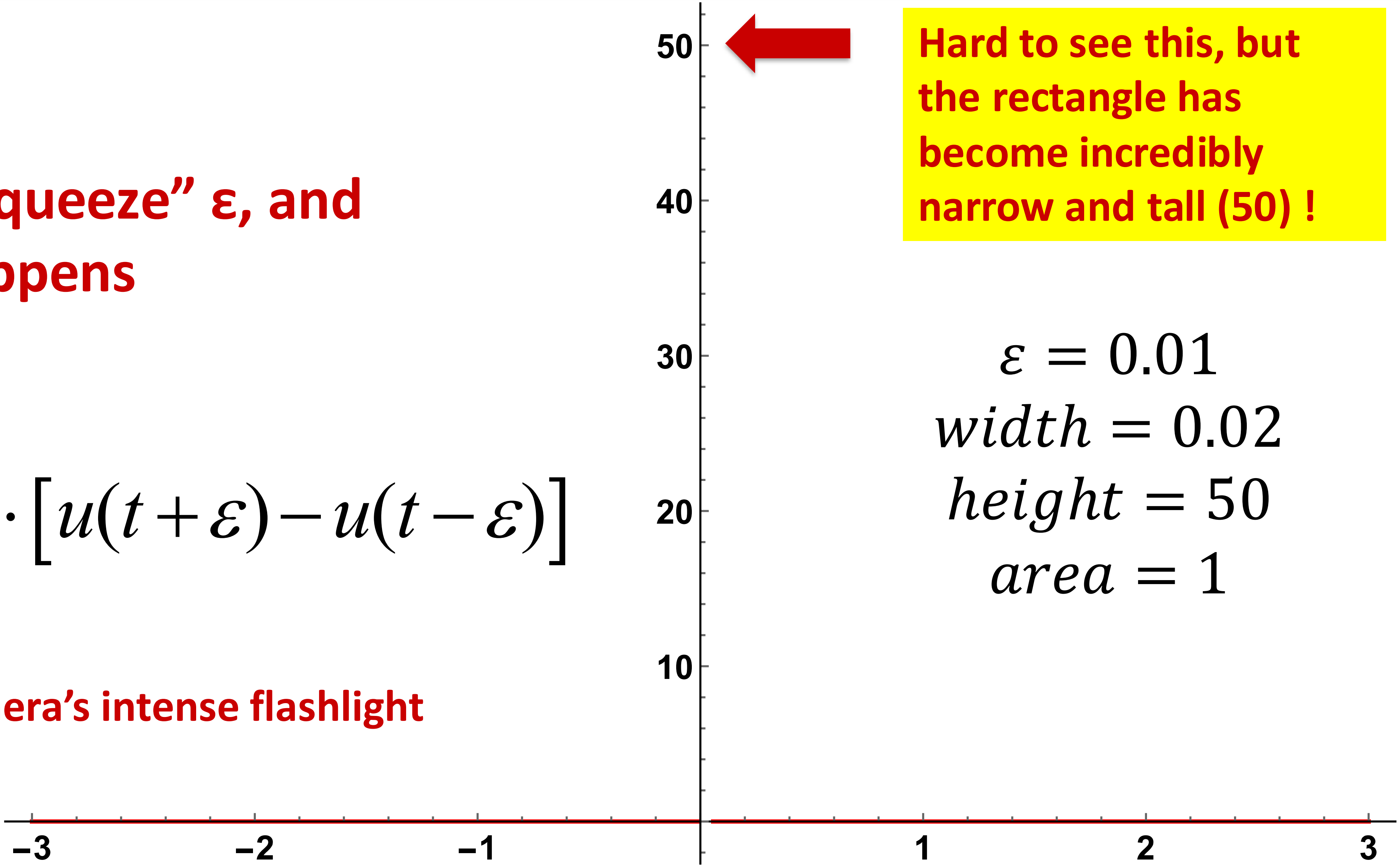


The Rectangle with Fixed Area

Now, let's "squeeze" ε, and see what happens

$$\Delta(t) = \frac{1}{2\varepsilon} \cdot \left[u(t+\varepsilon) - u(t-\varepsilon) \right]$$

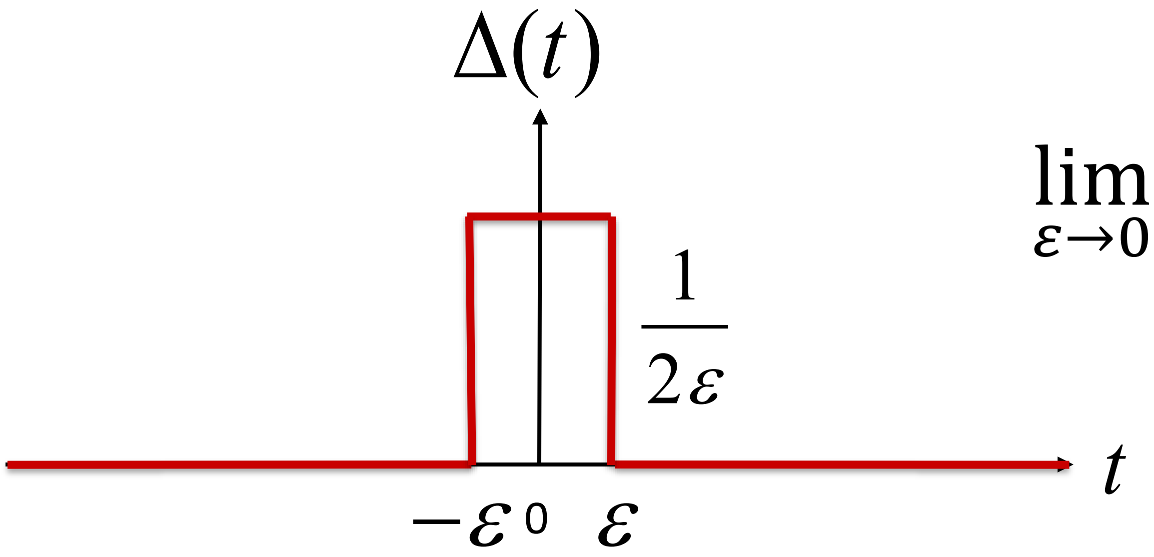
Analogy: A camera's intense flashlight



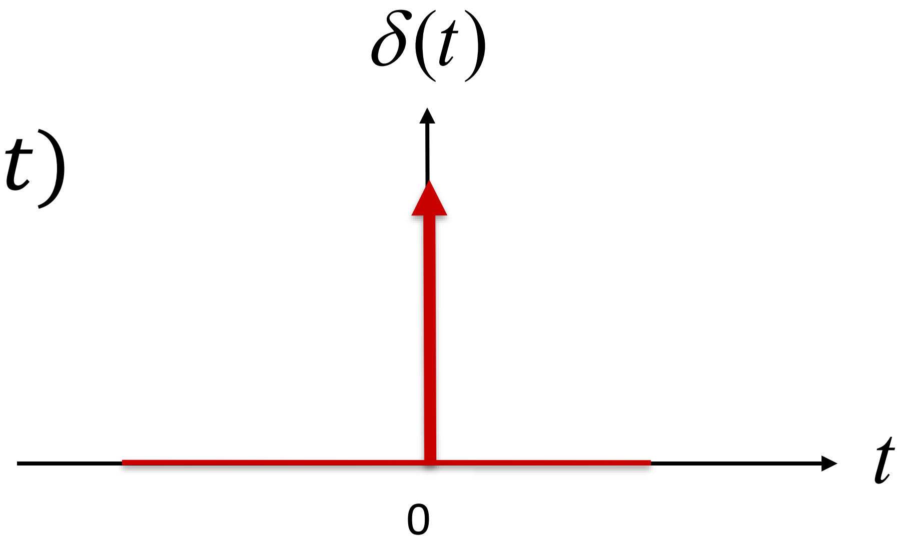
Hard to see this, but the rectangle has become incredibly narrow and tall (50) !

$$\varepsilon = 0.01$$
$$width = 0.02$$
$$height = 50$$
$$area = 1$$

The Limiting Case



$$\lim_{\varepsilon \to 0} \Delta(t) = \delta(t)$$

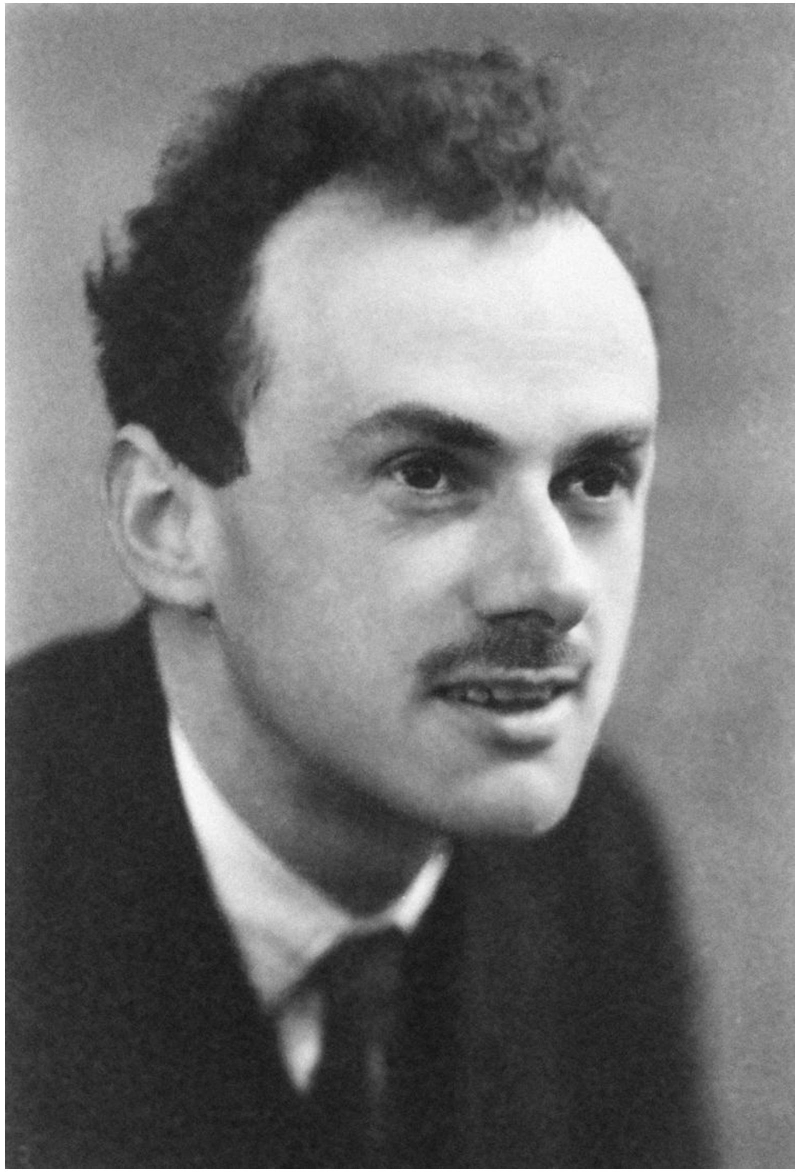


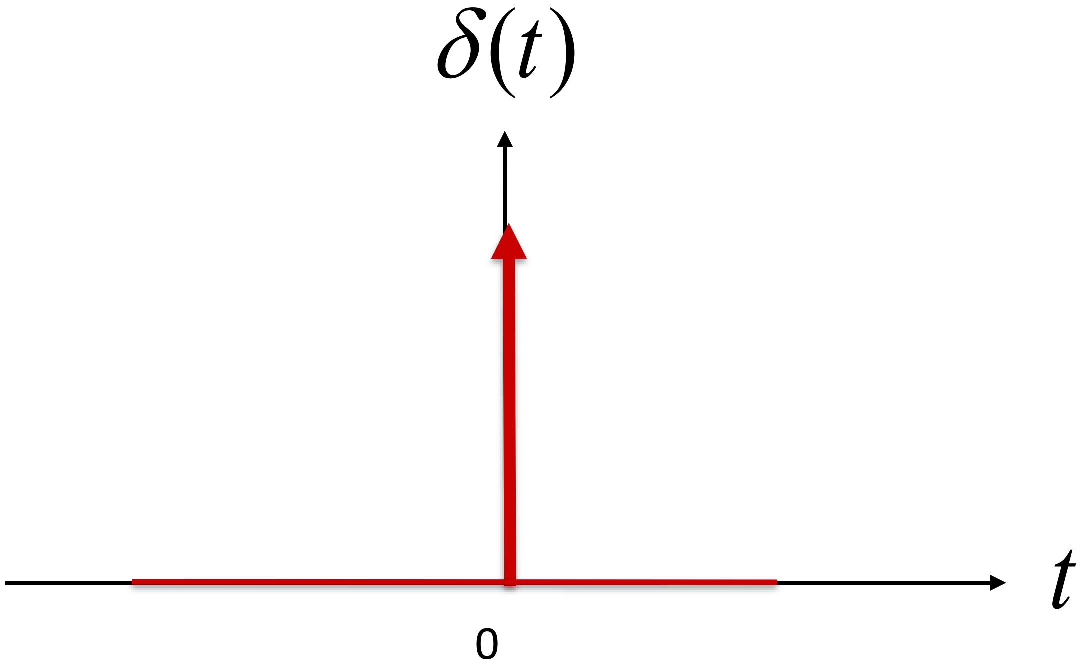
As $\varepsilon \to 0$ the limiting case $\delta(t)$ has “zero” width, and infinite height at 0

<u>But it still has an area (integral) of 1</u>

$$\int_{-\infty}^{+\infty} \delta(t)dt = 1$$

Physics Genius Made Mathematicians Cringe



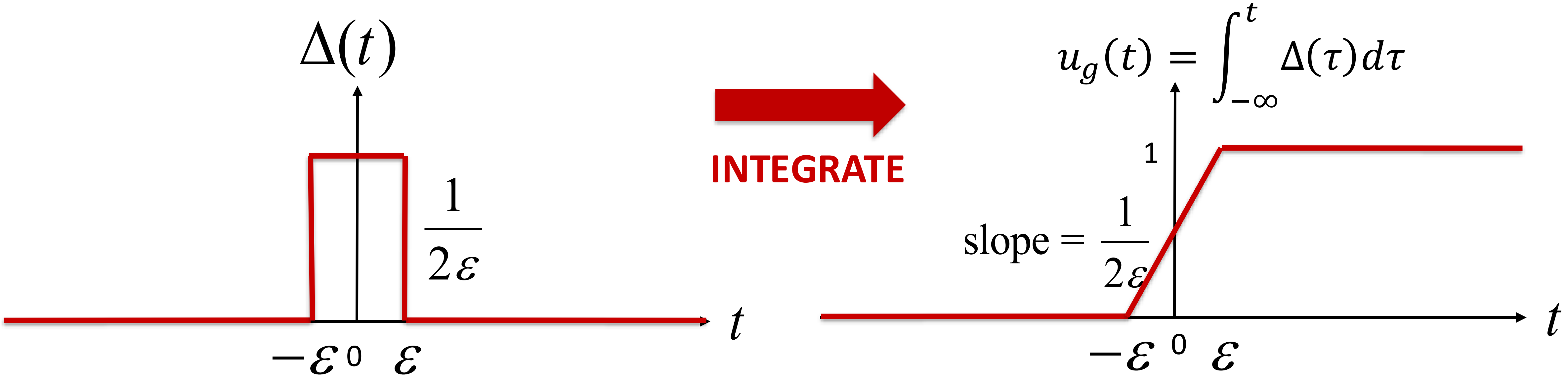


- $\delta(t)$ is known as the Impulse Function
- Also called the "Dirac Delta" function
- Named in honor of physicist Paul Dirac (1902 – 1984)

The Step Function as a Limiting Case

Let's integrate the Δ(t) function:

The Gradual Step Function



In the limit $\varepsilon \to 0$, $u_g(t)$ approaches the unit step $u(t)$

$$u(t) = \lim_{\varepsilon \to 0} u_g(t)$$

Basic Properties of the Impulse

We can actually define the impulse using the following two statements:

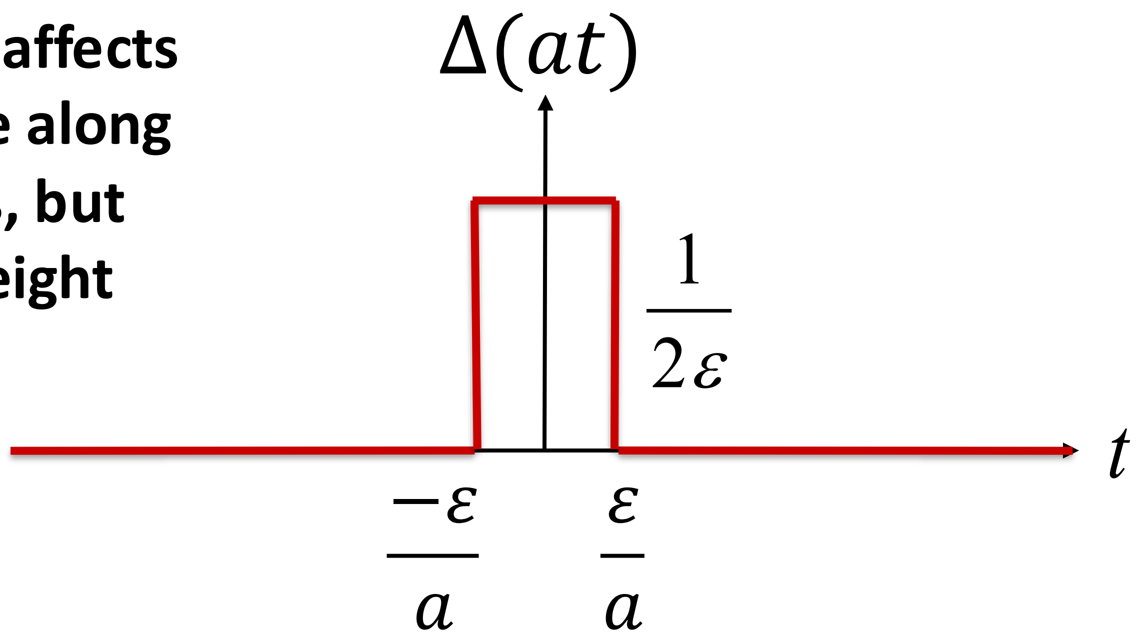
$$\delta(t-T)=0 \quad t \neq T \qquad\qquad \int_{-\infty}^{\infty} \delta(t-T)\,dt=1$$

Relationship to the step function:

$$\frac{du(t)}{dt}=\delta(t) \qquad\qquad u(t)=\int_{-\infty}^{t} \delta(\tau)\,d\tau$$

Time Scaling the Impulse Function

Time scaling affects the rectangle along the time axis, but leaves the height unchanged



$$\text{width} = \frac{2\varepsilon}{|a|}$$

$$\text{height} = \frac{1}{2\varepsilon}$$

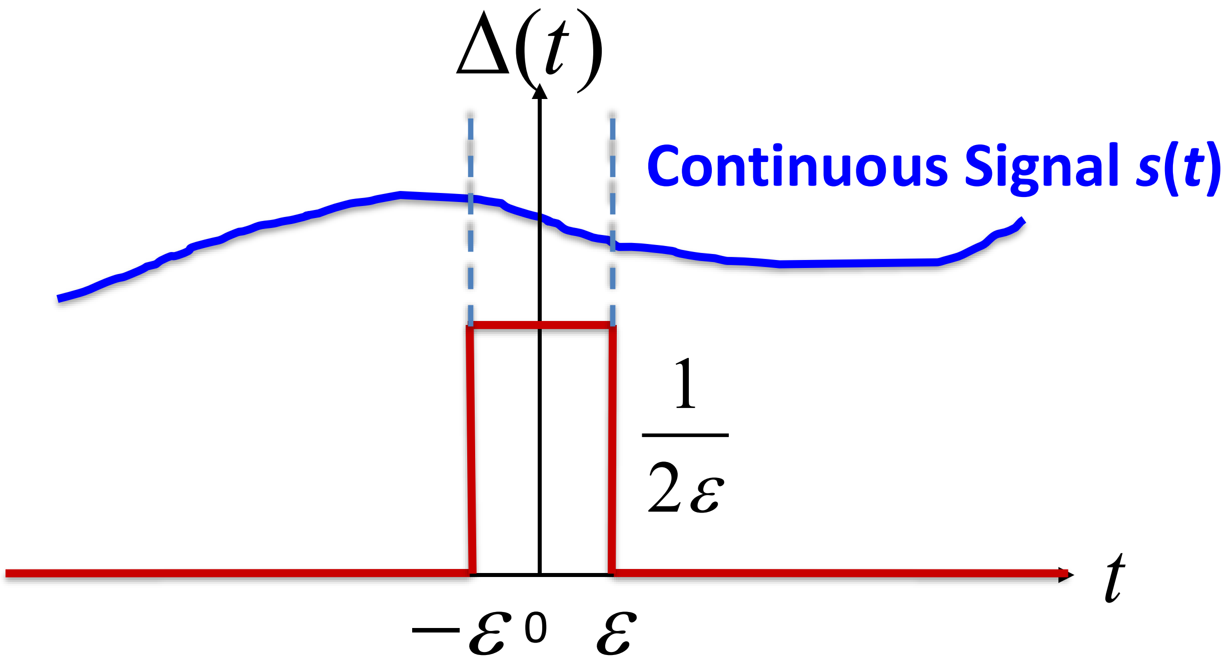
$$\text{area} = \frac{2\varepsilon}{|a|} \square \frac{1}{2\varepsilon} = \frac{1}{|a|}$$

$$\text{Therefore, } \delta(at) = \frac{1}{|a|}\delta(t)$$

Question:

$$\delta(-t) = ?$$

Taking a Snapshot of a Signal



Basically, $\Delta(t)s(t)$ acts like a "gate" that only keeps the part of the signal $s(t)$ between $-\varepsilon$ and $+\varepsilon$, and zeroes out everything else

Multiply $s(t)$ by $\Delta(t)$

$$\int_{-\infty}^{+\infty} \Delta(t)s(t)dt = \int_{-\varepsilon}^{+\varepsilon} \Delta(t)s(t)dt$$

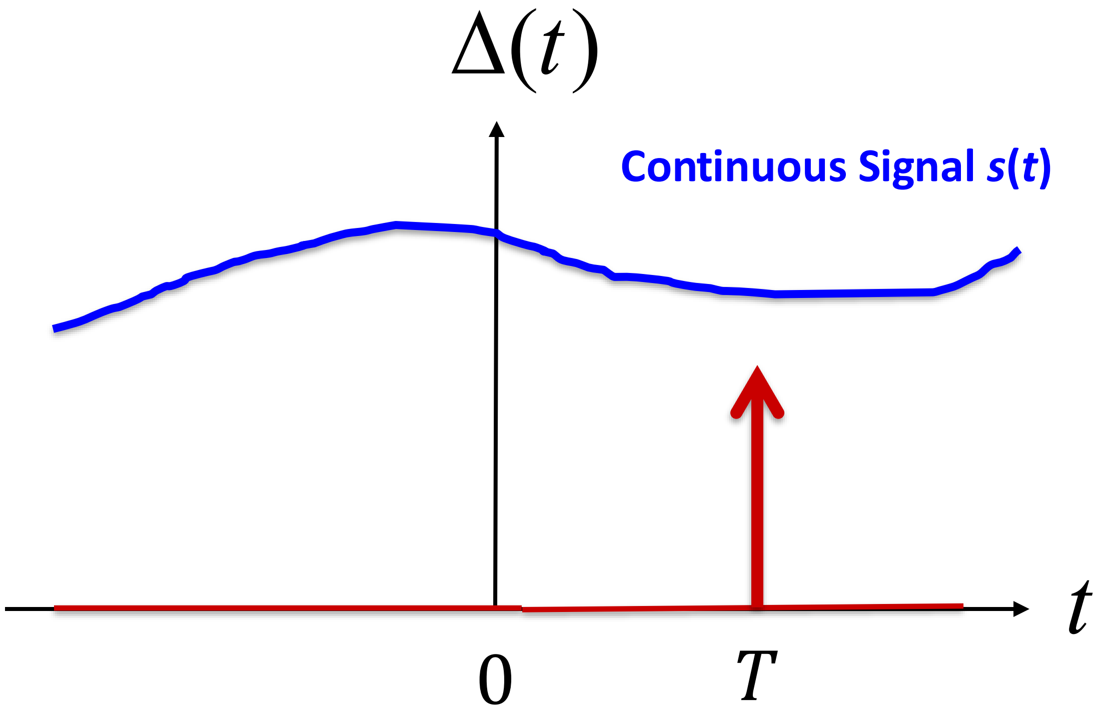
$$\approx 2\varepsilon \cdot \frac{1}{2\varepsilon} \cdot s(0)$$

In the Limit as $\varepsilon \to 0$, this is $s(0)$

$$\int_{-\infty}^{+\infty} \delta(t)s(t)dt = s(0)$$

SAMPLING PROPERTY

The Limiting Case



We can take the snapshot of a signal at a chosen time T, using a shifted delta function.

$$\int_{-\infty}^{+\infty} s(t)\cdot\delta(t-T)\,dt = s(T)$$

The snapshot is taken at $t = T$, the spot where $\delta(t-T) \neq 0$

Remember: This only works when the signal is continuous at T, when there is no sudden jump at the sampling time T

Calculus With the Impulse

When any signal $x(t)$ is multiplied by the impulse $\delta(t)$, the impulse zeroes out all values of the signal except when $t = 0$, so

$$x(t)\delta(t) = x(0)\delta(t)$$

Example 1:

VARIABLE

$$\int_{-\infty}^{t} x(\tau)\delta(\tau)d\tau = \int_{-\infty}^{t} x(0)\delta(\tau)d\tau = x(0)\int_{-\infty}^{t} \delta(\tau)d\tau = x(0)u(t)$$

FUNCTION

Example 2:

CONSTANT

$$\int_{-1}^{1} x(\tau)\delta(\tau)d\tau = \int_{-1}^{1} x(0)\delta(\tau)d\tau = x(0)\int_{-1}^{1} \delta(\tau)d\tau = x(0)\cdot 1 = x(0)$$

CONSTANT

Example

Evaluate the following integrals:

$$I_1 = \int_{-10}^{+10} t^2 \delta(t-3)dt = 3^2 = 9$$ **Snapshot of t^2 at 3**

$$I_2 = \int_{-1}^{+1} t^2 \delta(t-3)dt = 0$$

Sampling point at 3 is out of Range !

Another Example

Evaluate the integral:

$$I = \int_1^2 t^2 \delta(2t-3)\,dt$$

Start by time scaling the impulse: $\delta(2t-3) = \delta\left(2(t - {}^3/_2)\right) = \frac{1}{2}\delta(t - {}^3/_2)$

$$I = \frac{1}{2}\int_1^2 t^2 \delta(t - {}^3/_2)\,dt = \frac{1}{2}\underline{\left(\frac{3}{2}\right)^2}$$

Snapshot of t^2 at 3/2

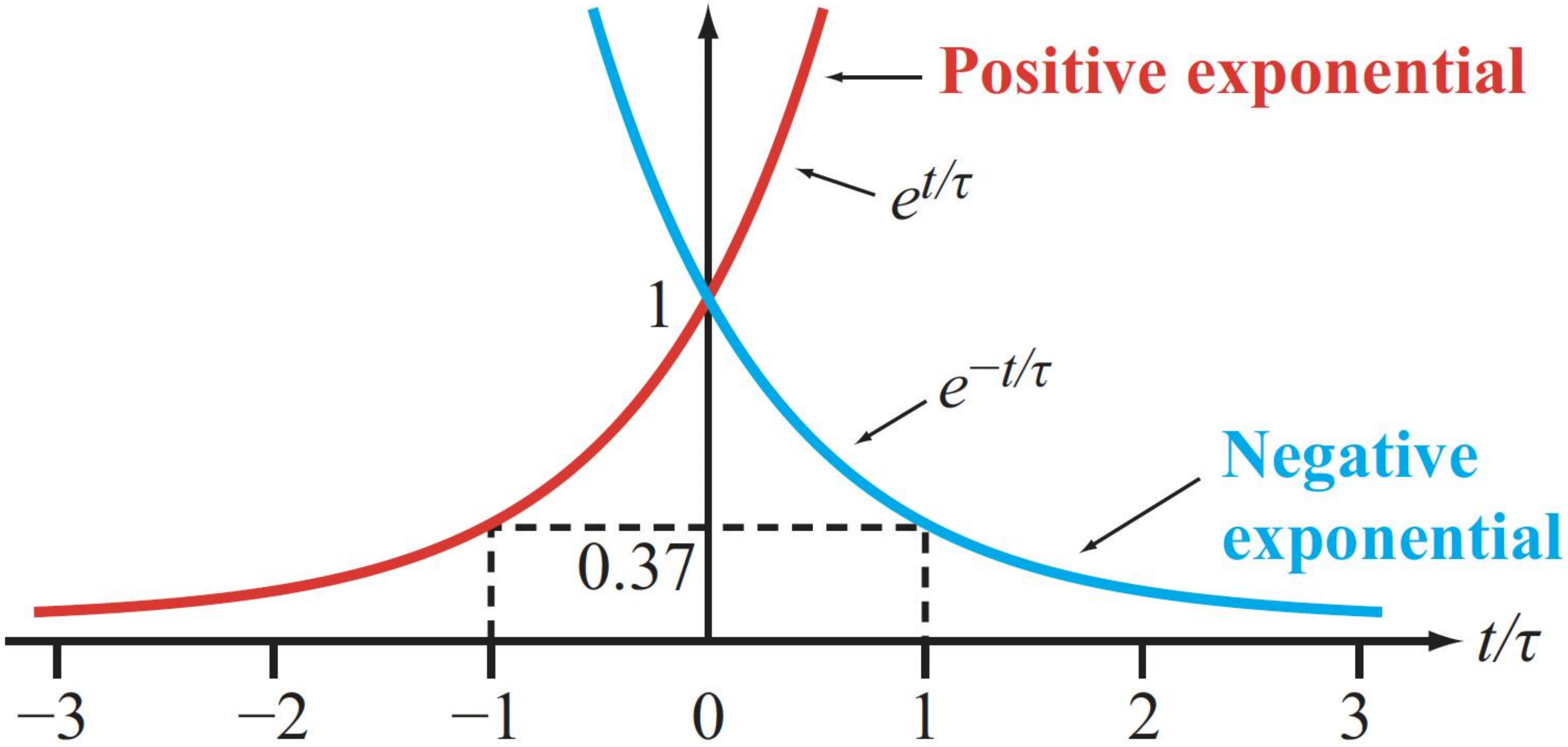
Aside: You can also do this problem by a change of variable $u = 2t - 3$

Exponential Signals

$$x_1(t) = e^{t/\tau}$$

$$x_2(t) = e^{-t/\tau}$$

$\tau =$ Time Constant



Exponential signals are very common, but we need a way to deal with the fact that real signals cannot be infinite in amplitude

Another Signal Property

▶ A signal $x(t)$ is:

- **causal** if $x(t) = 0$ for $t < 0$ (starts at or after $t = 0$)
- **noncausal** if $x(t) \neq 0$ for any $t < 0$ (starts before $t = 0$)
- **anticausal** if $x(t) = 0$ for $t > 0$ (ends at or before $t = 0$) ◀

Multiplying any signal by the unit step results in a causal signal

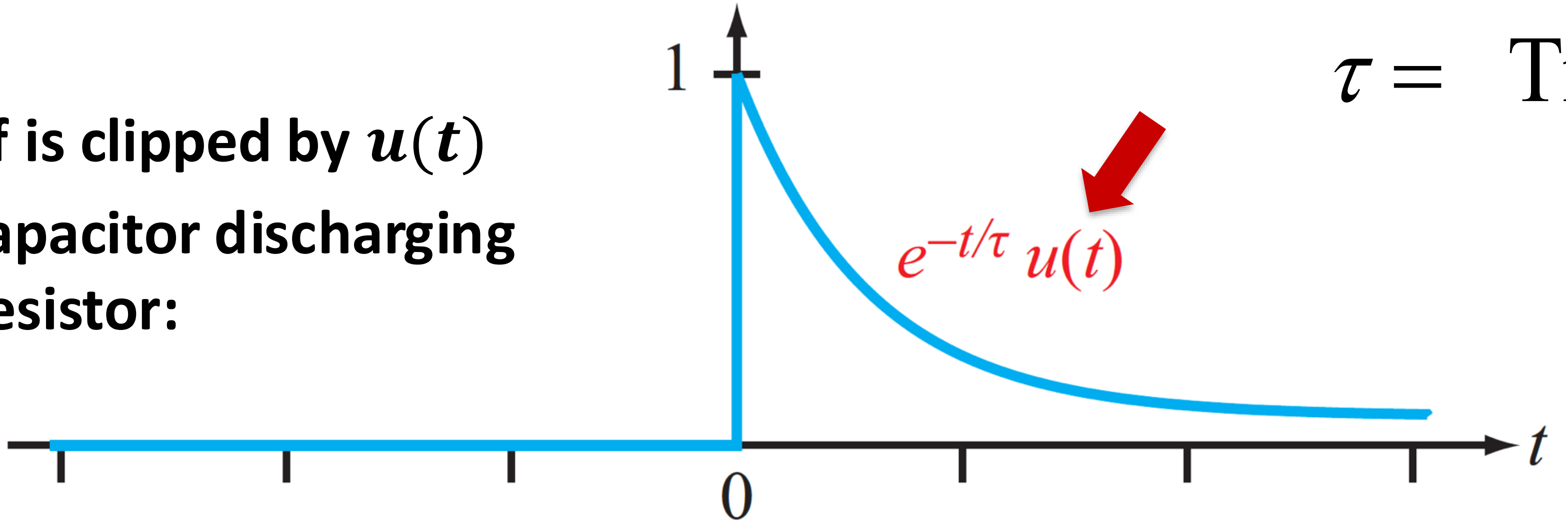
The unit step can be used as a “causality enforcer”

Exponential Signals "Gated/Clipped" by $u(t)$

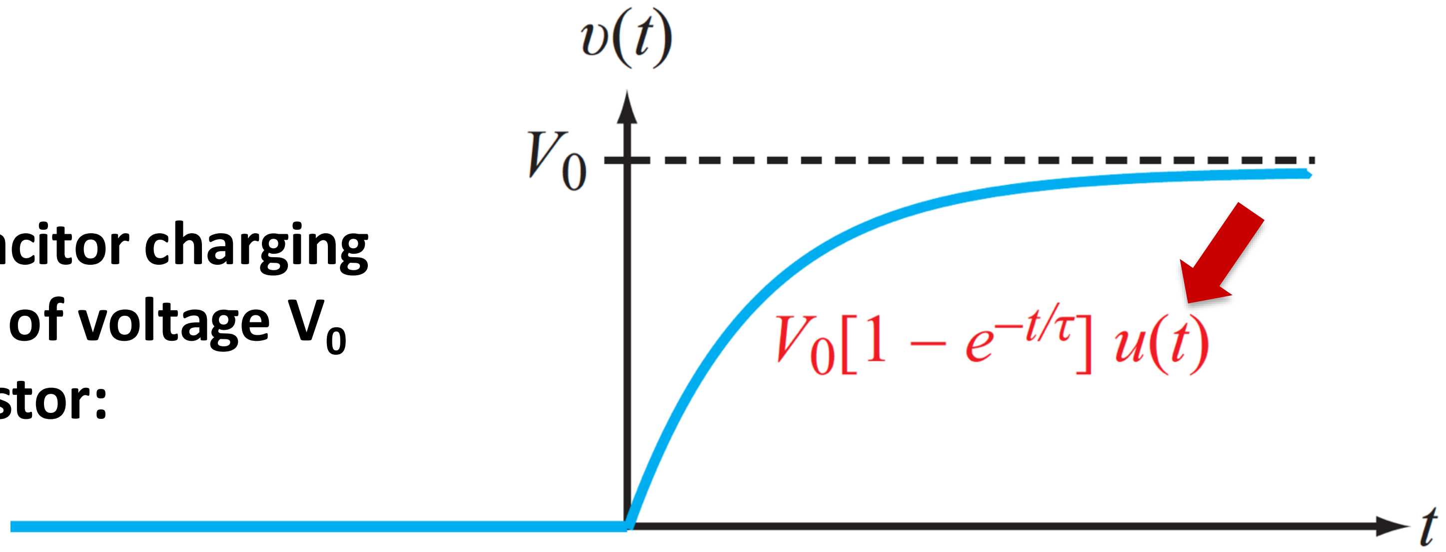
$\tau =$ Time Constant

The left half is clipped by $u(t)$

Example: Capacitor discharging through a resistor:



Example: Capacitor charging with a battery of voltage V_0 through a resistor:



Waveform Synthesis

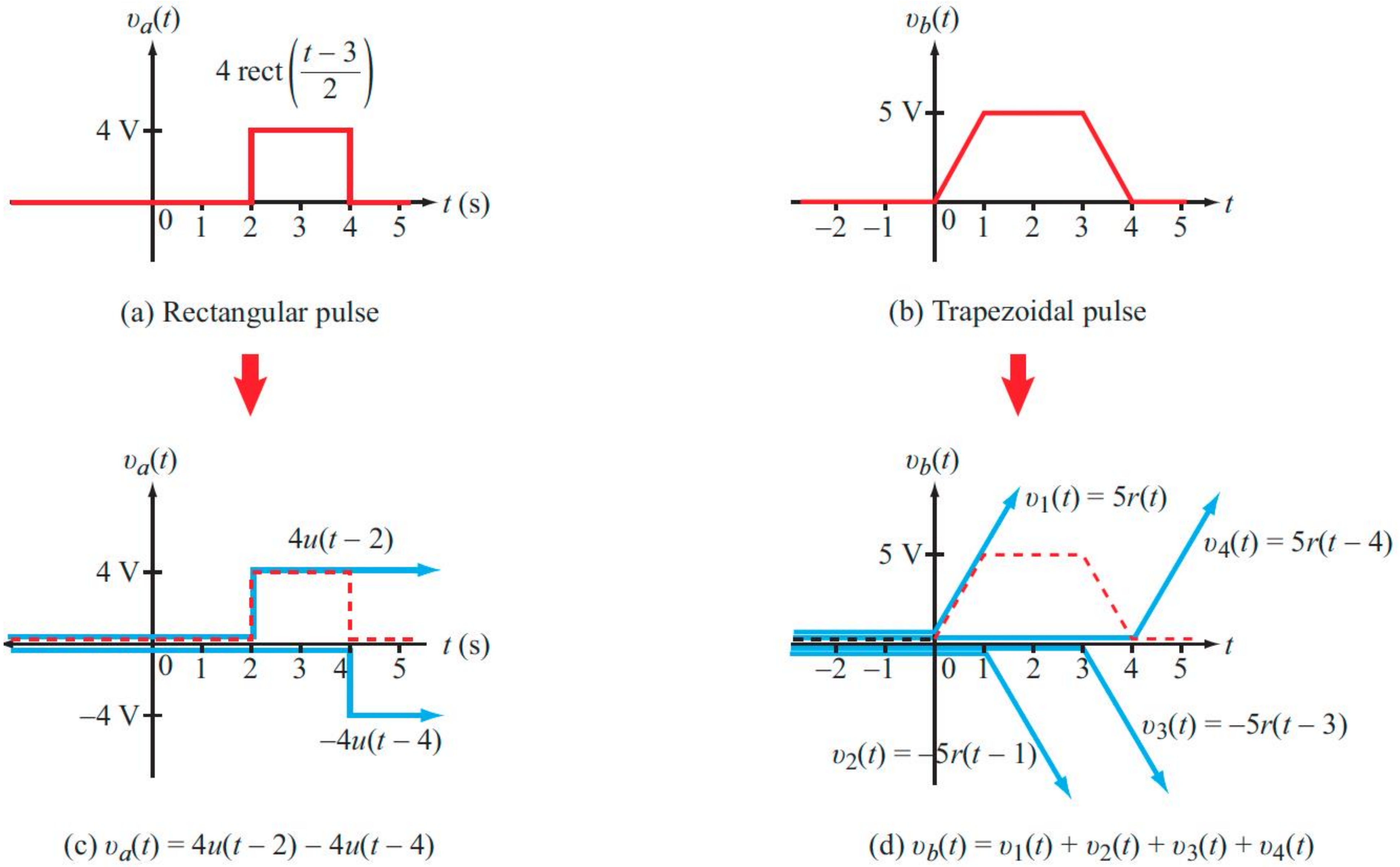
Thus far, we have learned:

- several fundamental waveforms (step, impulse, ramp, exponential, etc.), and
- basic ways to transform waveform (time shift, time scaling, amplitude scaling, etc.)

Using what we have learned, we can make up new waveforms

- We call this “waveform synthesis”

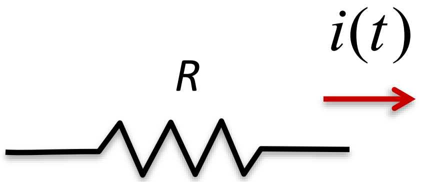
Examples



$v_a(t)$
$4 \text{ rect}\left(\frac{t-3}{2}\right)$
4 V
0 1 2 3 4 5
t (s)
(a) Rectangular pulse
$v_b(t)$
5 V
−2 −1 0 1 2 3 4 5
t
(b) Trapezoidal pulse
$v_a(t)$
$4u(t-2)$
4 V
−4 V
$-4u(t-4)$
0 1 2 3 4 5
t (s)
(c) $v_a(t) = 4u(t-2) - 4u(t-4)$
$v_b(t)$
$v_1(t) = 5r(t)$
$v_4(t) = 5r(t-4)$
5 V
−2 −1 0 1 2 3 4 5
t
$v_3(t) = -5r(t-3)$
$v_2(t) = -5r(t-1)$
(d) $v_b(t) = v_1(t) + v_2(t) + v_3(t) + v_4(t)$

Signal Power & Energy

Resistor heating



$$p(t) = i^2(t) \cdot R$$

Watts

Analogously, for <u>any</u> signal, we define:

$$\text{Instantaneous Power} = p(t) = |x(t)|^2$$

For complex signals, remember that $p(t) = x(t)x^*(t)$,

where $x^*(t)$ is the complex conjugate

In reality, a signal may not represent a physical thing like current or voltage

***Nevertheless,* we find it useful to define the notion of signal power**

Signal Power & Energy

$$\text{Instantaneous Power} = p(t) = |x(t)|^2$$

Think of power as yet another signal

Energy expended from time t_1 to t_2 is

$$E = \int_{t_1}^{t_2} p(t)\, dt$$

Total energy expended is

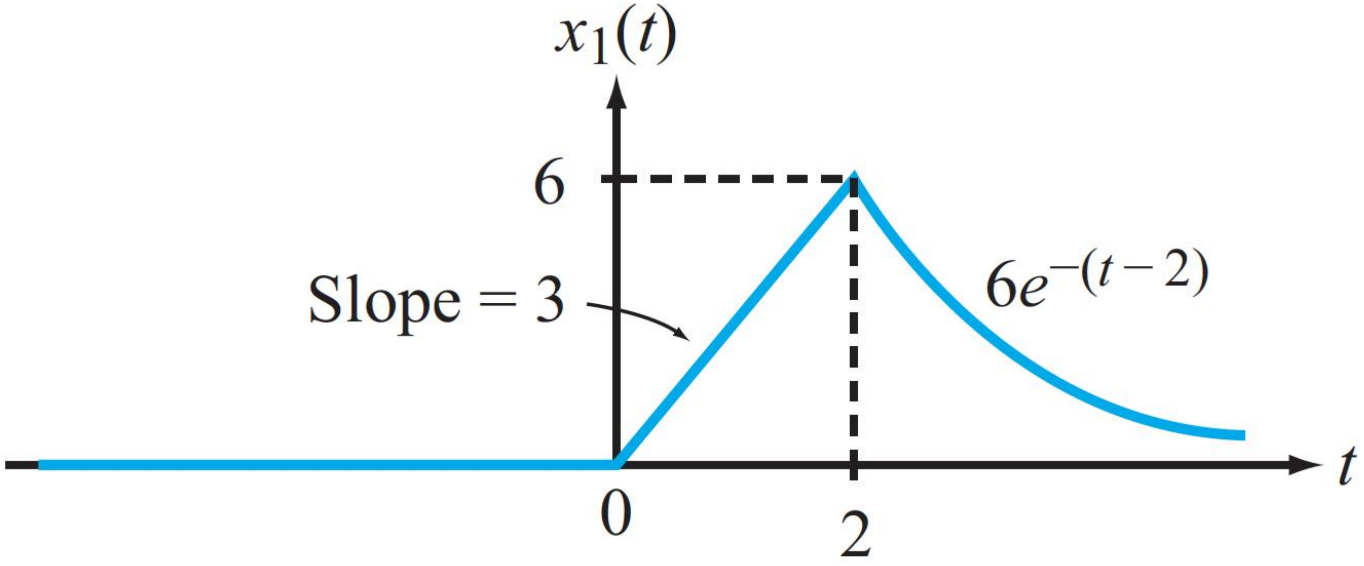
$$E = \int_{-\infty}^{\infty} p(t)\, dt$$

Average Power

$$P_{av} = \lim_{T \to \infty} \frac{1}{T} \int_{-T/2}^{+T/2} p(t)\, dt = \lim_{T \to \infty} \frac{1}{T} \int_{-T/2}^{+T/2} |x(t)|^2\, dt$$

Example 1

Calculate the power and energy for the following signal:



$$x_1(t) = \begin{cases} 0 & t \le 0 \\ 3t & 0 \le t \le 2 \\ 6e^{-(t-2)} & t \ge 2 \end{cases}$$

Example 1 (contd)

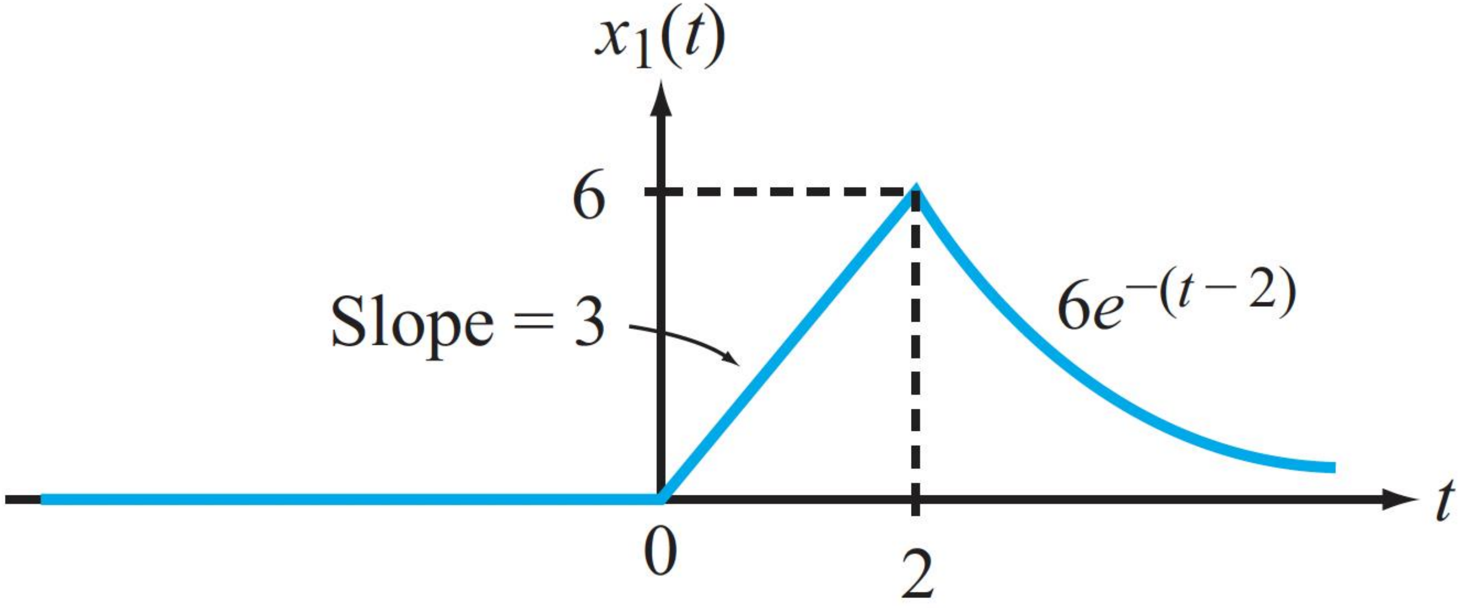
$$E_1 = \int_0^2 (3t)^2\,dt + \int_2^\infty [6e^{-(t-2)}]^2\,dt$$

$$= \int_0^2 9t^2\,dt + \int_2^\infty 36e^{-2(t-2)}\,dt$$

$$= \left.\frac{9t^3}{3}\right|_0^2 + 36e^4 \int_2^\infty e^{-2t}\,dt$$

$$= 24 + 36e^4 \left(\left.\frac{-e^{-2t}}{2}\right|_2^\infty \right)$$

$$= 42.$$



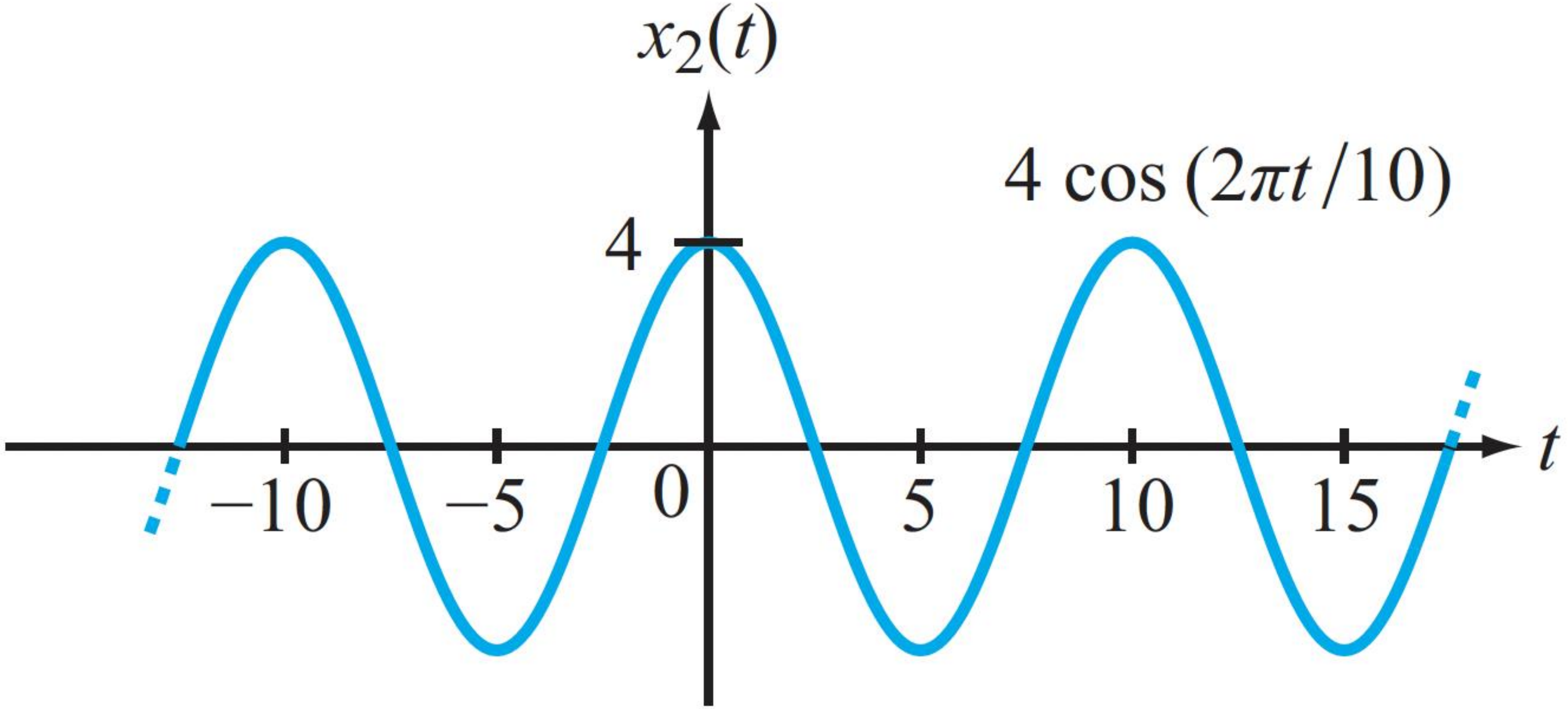
Eq. 1.35: If E is finite, then:

$$P_{av} = \lim_{T \to \infty} \frac{E}{T} = 0$$

Since E_1 is finite, it follows from Eq. (1.35) that $P_{\text{av}_1} = 0$.

Example 2

Calculate the power and energy for the following signal:



$$x_2(t) = 4\cos\left(\frac{2\pi t}{10}\right)$$

Example 2 (contd)

$$x_2(t) = 4\cos\left(\frac{2\pi t}{10}\right).$$

From the argument of $\cos(2\pi t/10)$, the period is 10 s. Hence, application of Eq. (1.36) leads to

$$P_{\text{av}_2} = \frac{1}{10}\int_{-5}^{5}\left[4\cos\left(\frac{2\pi t}{10}\right)\right]^2 dt$$

$$= \frac{1}{10}\int_{-5}^{5} 16\cos^2\left(\frac{2\pi t}{10}\right) dt$$

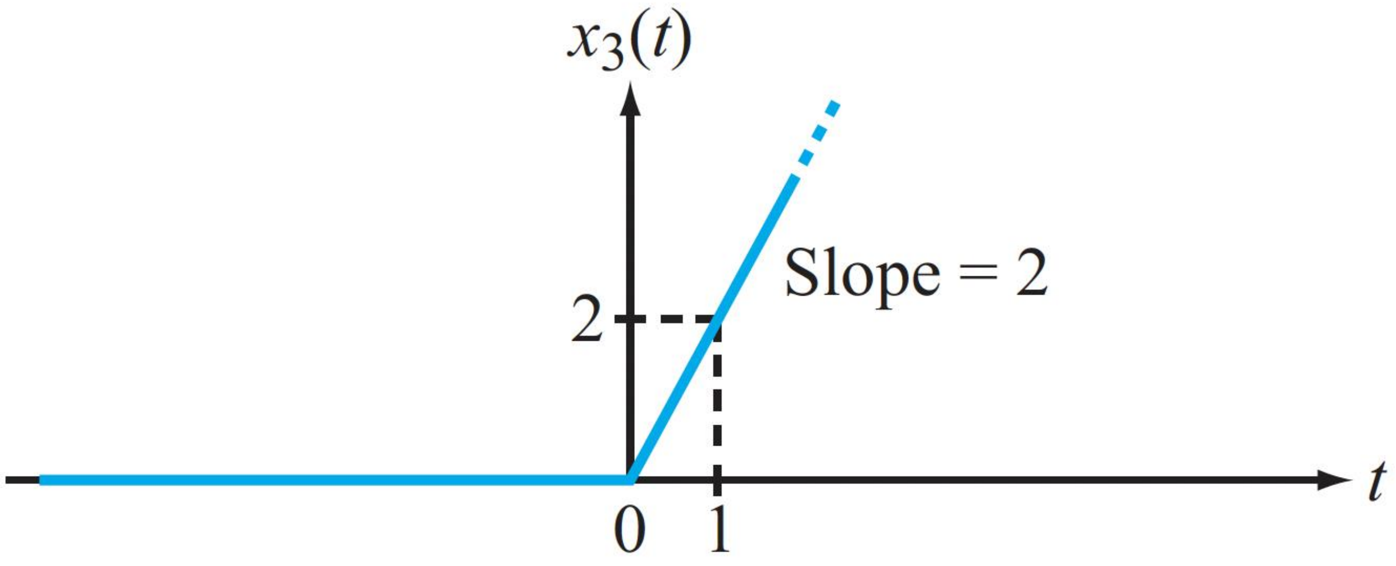
$$= 8.$$

Since P_{av2} is finite, total E_2 is infinite.

Example 3

The time-averaged power associated with $x_3(t)$ is

$$
\begin{aligned}
P_{\mathrm{av}_3} &= \lim_{T\to\infty} \frac{1}{T} \int_0^{T/2} 4t^2\, dt \\
&= \lim_{T\to\infty} \frac{1}{T} \left[\left. \frac{4t^3}{3} \right|_0^{T/2} \right] \\
&= \lim_{T\to\infty} \left[\frac{1}{T} \times \frac{4T^3}{24} \right] \\
&= \lim_{T\to\infty} \left[\frac{T^2}{6} \right] \to \infty.
\end{aligned}
$$

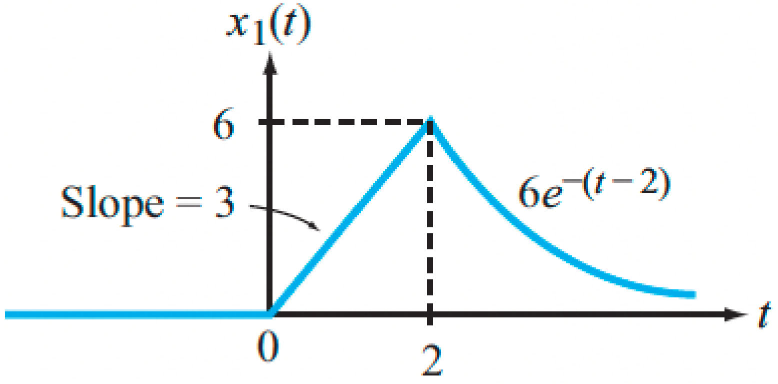


Moreover, $E_3 \to \infty$ as well.

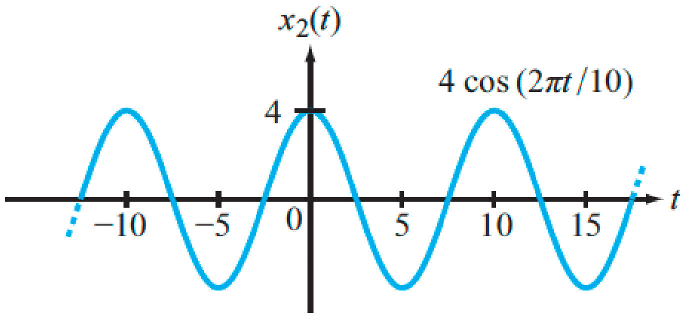
Three Types of Signals

Based on their power and energy profiles, there are 3 types of signals:

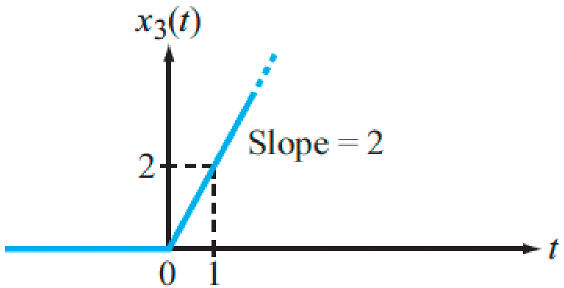
Energy signals: $P_{av} = 0$ and E is finite



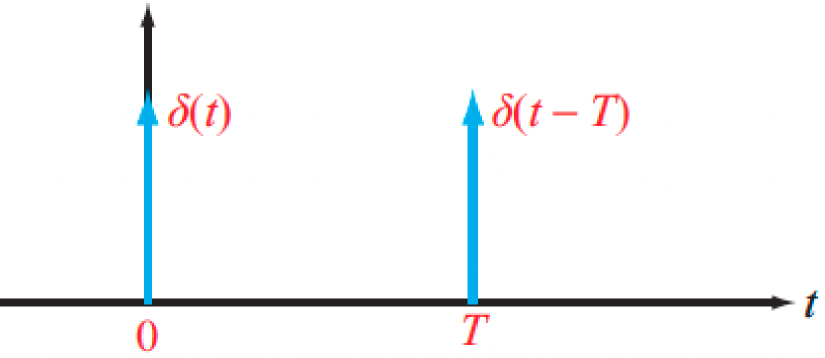
Power signals: P_{av} is finite and E is infinite

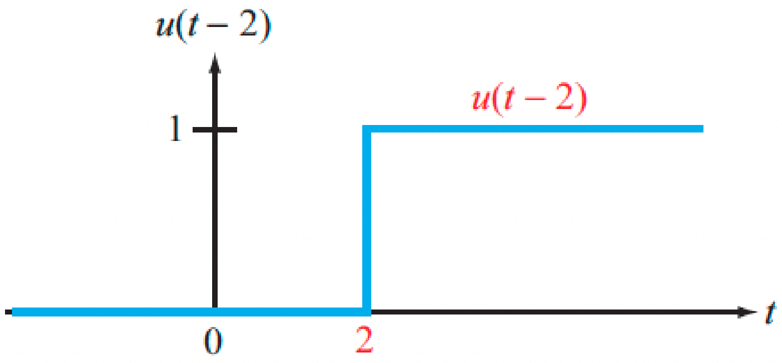


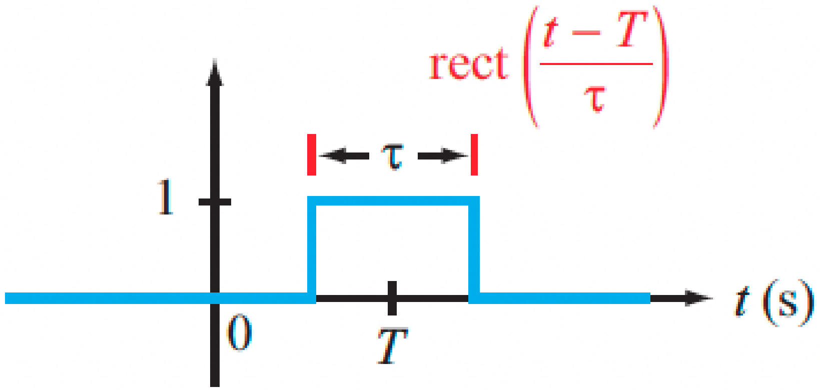
Non-physical signals: P_{av} and E are infinite

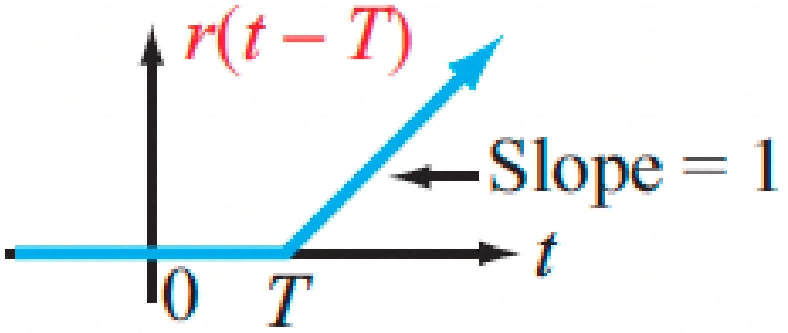


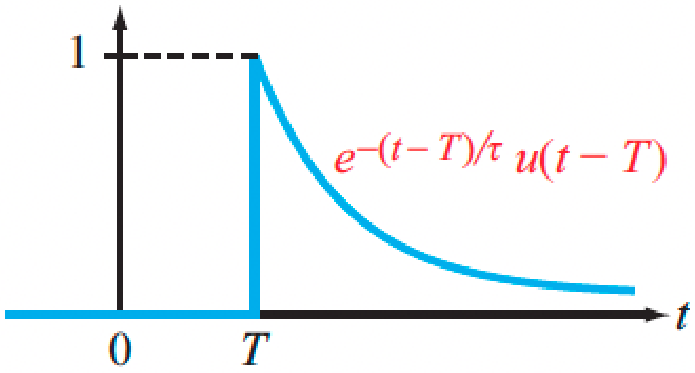
Summary of Basic Waveforms

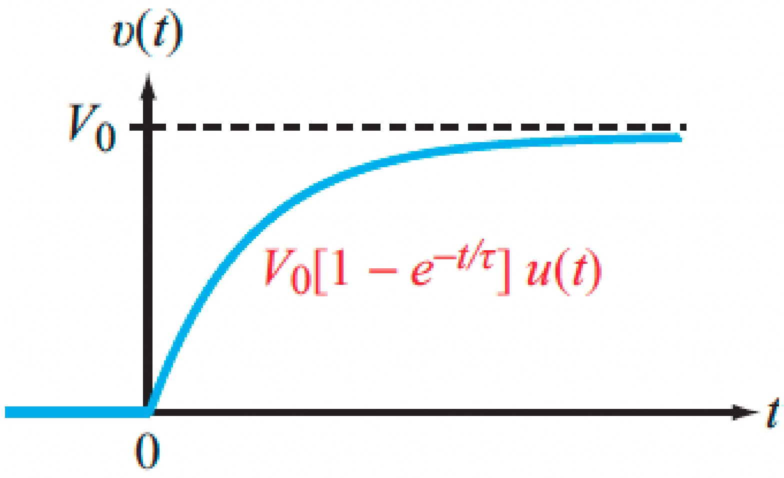












For the Next Class

Work through the PDF textbook examples in Sections 1-4.4 – 1-5

Study Sections 2-1 – 2.2 (New Chapter!) on Zybooks

How to Study:

1. Read the assigned sections in order to understand the basic concepts, and aim to memorize the concepts
2. Attempt the in-chapter exercises and practice until you can do them fully on your own without notes, like in a test
3. Write down your questions and bring them to class